

# **Customer Satisfaction Clustering and Sales Forecasting Analysis**

## **Using the Olist Brazilian E-Commerce Dataset**

### **Team Brazil Retail Analytics**

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# Table of Contents

|                                                                                                                                                                                                                                                       |    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1. Abstract / Executive Summary.....                                                                                                                                                                                                                  | 9  |
| 2. Project Plan .....                                                                                                                                                                                                                                 | 11 |
| Business Description .....                                                                                                                                                                                                                            | 11 |
| 1. Commerce Solutions .....                                                                                                                                                                                                                           | 12 |
| 2. Logistics (Olist Pax) .....                                                                                                                                                                                                                        | 12 |
| 3. Financial Services (Olist Credit & Olist Pay) .....                                                                                                                                                                                                | 12 |
| Who Olist Serves .....                                                                                                                                                                                                                                | 12 |
| Financials.....                                                                                                                                                                                                                                       | 12 |
| Key Executives .....                                                                                                                                                                                                                                  | 13 |
| Major Competitors .....                                                                                                                                                                                                                               | 13 |
| 2.3 Business and Analysis Opportunity .....                                                                                                                                                                                                           | 13 |
| 2.4 Research Questions and Sub-Questions.....                                                                                                                                                                                                         | 15 |
| RQ1 Sub-Questions .....                                                                                                                                                                                                                               | 15 |
| RQ2 Sub-Questions .....                                                                                                                                                                                                                               | 15 |
| 2.5 Hypotheses and Planned Analytical Techniques .....                                                                                                                                                                                                | 16 |
| H1: Products and orders belonging to clusters with shorter delivery times, lower freight costs, favorable product categories, and stronger operational performance will exhibit significantly higher customer review scores than other clusters. .... | 16 |
| Analytical Techniques: .....                                                                                                                                                                                                                          | 16 |
| H2. Historical sales transaction patterns contain sufficient seasonal and trend information to accurately forecast future sales revenue and order demand. ....                                                                                        | 16 |

|                                                          |    |
|----------------------------------------------------------|----|
| Analytical Techniques: .....                             | 16 |
| 2.6 Dataset Source.....                                  | 17 |
| Dataset Information .....                                | 17 |
| 2.7 Measurements .....                                   | 17 |
| 2.8 Planned Methodology .....                            | 18 |
| 2.9 Computational Methods and Outputs.....               | 19 |
| 2.10 Output Summaries.....                               | 20 |
| Output 1 – Customer Satisfaction Cluster Profiles.....   | 20 |
| Output 2 – Customer Satisfaction Driver Analysis .....   | 20 |
| Output 3 – Sales Forecasting Dashboard.....              | 21 |
| 2.11 Campaign Implementation .....                       | 21 |
| 3. Literature Review.....                                | 22 |
| 3.1 Introduction .....                                   | 22 |
| 3.2 Customer Satisfaction in E-Commerce.....             | 22 |
| 3.3 Operational Drivers of Customer Satisfaction .....   | 23 |
| 3.4 Customer Segmentation and Clustering Techniques..... | 24 |
| 3.5 Sales Forecasting in E-Commerce .....                | 24 |
| 3.6 Research Gap .....                                   | 25 |
| 3.7 Conclusion.....                                      | 26 |
| 4. Final Research Questions.....                         | 27 |
| 5. Exploratory Data Analysis .....                       | 28 |
| 5.1 Introduction .....                                   | 28 |
| 5.2 Dataset Overview .....                               | 28 |
| Dataset Inventory.....                                   | 29 |

|                                                                   |    |
|-------------------------------------------------------------------|----|
| 5.3 Data Quality Assessment.....                                  | 29 |
| Missing-Value Analysis .....                                      | 29 |
| Major Missing Values Identified .....                             | 30 |
| Duplicate Record Analysis .....                                   | 30 |
| Data-Type Validation.....                                         | 30 |
| 5.4 Feature Engineering .....                                     | 31 |
| 5.5 Customer Satisfaction and Operational Performance (RQ1) ..... | 31 |
| Analysis Overview .....                                           | 31 |
| Descriptive Statistics .....                                      | 32 |
| Summary Statistics .....                                          | 32 |
| Review Score Distribution .....                                   | 33 |
| Distribution of Customer Review Scores .....                      | 33 |
| Product Price Distribution .....                                  | 34 |
| Distribution of Product Prices .....                              | 34 |
| Freight Value Distribution .....                                  | 35 |
| Distribution of Freight Values .....                              | 35 |
| Payment Value Distribution.....                                   | 36 |
| Distribution of Payment Values.....                               | 36 |
| Delivery-Performance Analysis.....                                | 37 |
| Distribution of Delivery Duration .....                           | 37 |
| Distribution of Delivery Delay .....                              | 38 |
| Outlier Analysis .....                                            | 38 |

|                                                                                                                                              |    |
|----------------------------------------------------------------------------------------------------------------------------------------------|----|
| Outlier Summary .....                                                                                                                        | 39 |
| Boxplots of Key Variables .....                                                                                                              | 39 |
| Relationship Analysis .....                                                                                                                  | 42 |
| Review Score vs. Delivery Duration .....                                                                                                     | 42 |
| Review Score vs. Delivery Delay .....                                                                                                        | 43 |
| Review Score vs. Freight Value .....                                                                                                         | 44 |
| Correlation Analysis .....                                                                                                                   | 45 |
| Correlation Heatmap .....                                                                                                                    | 45 |
| Product-Category Analysis .....                                                                                                              | 46 |
| Top-Rated Product Categories .....                                                                                                           | 46 |
| 5.6 Sales Trends and Forecasting Exploration (RQ2) .....                                                                                     | 46 |
| Analysis Overview .....                                                                                                                      | 46 |
| Monthly Revenue Trend .....                                                                                                                  | 47 |
| Monthly Order-Volume Trend .....                                                                                                             | 48 |
| Moving-Average Analysis .....                                                                                                                | 49 |
| Monthly Revenue with Three-Month Moving Average .....                                                                                        | 49 |
| 5.7 EDA Findings, Limitations, and Modeling Implications .....                                                                               | 49 |
| 5.8 EDA Reconciliation Note .....                                                                                                            | 50 |
| 6. Methodology .....                                                                                                                         | 51 |
| 6.1 RQ1: Can clustering identify distinct groups of products/orders based on customer<br>satisfaction and operational characteristics? ..... | 51 |
| Data Preparation .....                                                                                                                       | 51 |

|                                                                                                  |    |
|--------------------------------------------------------------------------------------------------|----|
| Variable Selection .....                                                                         | 51 |
| Data Transformation .....                                                                        | 52 |
| Clustering Technique .....                                                                       | 53 |
| Cluster Profiling .....                                                                          | 53 |
| Statistical Validation .....                                                                     | 54 |
| Evaluation Metrics .....                                                                         | 54 |
| Completed RQ1 Implementation .....                                                               | 54 |
| 6.2 RQ2: How accurately can future sales trends be forecast using historical transaction data? . |    |
| 54                                                                                               |    |
| Data Preparation .....                                                                           | 54 |
| Variable Selection .....                                                                         | 55 |
| Time-Series Aggregation .....                                                                    | 55 |
| Forecasting Techniques .....                                                                     | 55 |
| Moving Average .....                                                                             | 56 |
| Exponential Smoothing .....                                                                      | 56 |
| ARIMA .....                                                                                      | 56 |
| Model Training and Validation .....                                                              | 56 |
| Evaluation Metrics .....                                                                         | 56 |
| Output and Interpretation .....                                                                  | 57 |
| Completed RQ2 Implementation .....                                                               | 57 |
| 7. Data Visualizations and Analysis .....                                                        | 58 |
| 7.1 Customer Satisfaction and Operational Performance .....                                      | 59 |
| Review Score Distribution .....                                                                  | 59 |

|                                                              |    |
|--------------------------------------------------------------|----|
| Delivery Performance.....                                    | 59 |
| Review Score by Delivery Performance .....                   | 60 |
| Correlation Matrix .....                                     | 60 |
| Product Category Review Scores .....                         | 61 |
| 7.2 K-Means Cluster Selection and Results .....              | 61 |
| Cluster Number Selection.....                                | 63 |
| 7.3 Cluster Profiles and Business Interpretation .....       | 64 |
| Cluster Sizes and Profiles.....                              | 64 |
| Key Variables by Cluster .....                               | 65 |
| PCA Cluster Visualization .....                              | 66 |
| 7.4 Historical Sales Trends.....                             | 66 |
| Monthly Revenue, Order Volume, and Average Order Value ..... | 67 |
| Revenue with Moving Average .....                            | 67 |
| 7.5 Forecasting Model Evaluation .....                       | 67 |
| Forecast Model Comparison.....                               | 68 |
| Forecast Accuracy .....                                      | 68 |
| 7.6 Overall Visualization Findings.....                      | 69 |
| 8. Ethical Recommendations .....                             | 71 |
| 8.1 Ethical Context.....                                     | 71 |
| 8.2 Customer Privacy and Responsible Data Use .....          | 71 |
| 8.3 Review-Score Bias and Missing Feedback .....             | 71 |
| 8.4 Seller and Regional Fairness in Delivery Decisions ..... | 72 |
| 8.5 Responsible Cluster Interpretation .....                 | 72 |
| 8.6 Forecasting, Inventory, and Product Diversity.....       | 73 |

|                                                               |    |
|---------------------------------------------------------------|----|
| 8.7 Regional and Category Fairness.....                       | 73 |
| 8.8 Risks of Ignoring the Findings.....                       | 73 |
| 8.9 Transparency, Human Oversight, and Responsible Use .....  | 73 |
| 9. Challenges and Limitations.....                            | 75 |
| 10. Recommendations and Next Steps.....                       | 77 |
| 10.1 Immediate Operational Recommendations .....              | 77 |
| 10.2 Customer-Satisfaction Recommendations .....              | 77 |
| 10.3 Delivery and Logistics Recommendations.....              | 77 |
| 10.4 Forecasting and Inventory-Planning Recommendations ..... | 77 |
| 10.5 Responsible-Use Safeguards .....                         | 78 |
| 10.6 Future Analytical Work .....                             | 78 |
| 11. Conclusion .....                                          | 79 |
| References.....                                               | 80 |
| Appendix A: Source Code .....                                 | 81 |
| Library Imports and Configuration .....                       | 81 |

## 1. Abstract / Executive Summary

In the highly competitive e-commerce landscape, sustained growth depends on an organization's ability to deliver consistent customer experiences while maintaining operational efficiency and anticipating future demand. For Olist, a leading Brazilian marketplace platform supporting thousands of small and medium-sized sellers, performance gaps in delivery reliability, logistics coordination, and demand forecasting directly impact customer satisfaction, seller success, and overall profitability. Addressing these challenges is essential not only for improving service quality but also for strengthening Olist's strategic positioning in a rapidly evolving digital marketplace.

This analysis was conducted to provide actionable, data-driven insights that support operational optimization and forward-looking decision-making. Using the Olist Brazilian E-Commerce Dataset, the study integrates customer experience analytics with sales forecasting to evaluate both current performance drivers and future revenue trends. The completed analytical dataset contains 113,425 order-item records representing 99,441 unique orders between 2016 and 2018, enabling a comprehensive assessment of transactional behavior, delivery performance, and customer feedback.

Two key business questions guided the analysis. First, whether distinct operational and customer satisfaction patterns can be identified through clustering techniques. Second, how effectively historical transaction data can be leveraged to forecast future sales performance. These questions were designed to align with core business priorities, including improving delivery reliability, enhancing customer retention, and strengthening demand planning capabilities.

To address the Research Q1, K-Means clustering was applied to six standardized variables: customer review score, delivery duration, delivery delay, item price, freight value, and payment value. Model diagnostics, including elbow and silhouette methods, supported the selection of a four-cluster solution that balances statistical validity with business interpretability. The results reveal a dominant high-performing segment (68.1%) characterized by strong customer satisfaction, efficient delivery timelines, and early fulfillment relative to estimated dates. In contrast, a delayed-delivery segment (9.5%) demonstrates extended delivery durations and lower operational performance, highlighting a critical area for improvement. Additional segments capture premium purchasing behavior and a subset of dissatisfied customers whose negative experi-

ences are not solely explained by delivery factors, indicating the presence of broader service or product-related issues.

For the Research Q2, multiple time series forecasting techniques were evaluated, including naïve forecasting, moving averages, simple exponential smoothing, and Holt's Linear Trend method. Using a five-month holdout validation approach, ARIMA (0,1,1) delivered the strongest forecasting performance, achieving an RMSE of R\$114,881 and a MAPE of 9.2%. Compared with the naïve baseline, ARIMA reduced RMSE by approximately 11%. These results indicate a strong capacity to capture underlying revenue trends. Further analysis shows that revenue growth during the observed period was primarily driven by increases in order volume rather than changes in average order value, suggesting that marketplace expansion and customer acquisition have been key growth drivers.

The outcomes of this analysis provide clear implications for Olist's operational and strategic functions. Logistics and operations teams can prioritize interventions targeting delayed deliveries, which are strongly associated with lower customer satisfaction. Seller management teams can leverage insights into high-performing transaction characteristics to guide seller support and performance improvement initiatives. Inventory planning and supply chain functions can utilize forecasting outputs to better align stock levels with anticipated demand, reducing inefficiencies and improving service levels. At the executive level, the integration of customer experience analytics with demand forecasting supports more informed decision-making, enabling Olist to balance growth objectives with service quality.

## 2. Project Plan

*The Project Plan below reflects the project's originally proposed scope. Completed procedures and results are documented in Sections 6 and 7.*

### 2.1 Team Members

- Team Lead – Marukha Hussaini
- Analyst – Anand Chamoli
- Analyst – Sofia Nagraj
- Analyst – Logan Lee

### 2.2 Organizational Profile and Background of the Opportunity

#### Primary Company Details

Founded - Feb 1, 2015

Founders - Tiago Dalvi

Headquarters - Curitiba, Paraná, Brazil

Categories - Home & Lifestyle, Health & Beauty, Electronics & Entertainment, Fashion, Leisure & Automotive

#### **Address:**

Avenida João Gualberto 1,698, Curitiba, Paraná, 80030-001, Brazil

#### **Company Communication:**

Website: [www.olist.com](http://www.olist.com)

Phone Number: +55 41 3203 9468

#### Business Description

Olist is a Brazilian e-commerce technology unicorn that acts as a "commerce enabler" for small and medium-sized businesses (SMBs). It provides a comprehensive SaaS ecosystem that helps retailers manage their digital operations, sell on major marketplaces, handle logistics, and secure business financing, all from one platform.

Olist operates across three primary business dimensions:

## ***1. Commerce Solutions***

Olist allows merchants to list and manage their products across major online marketplaces and social platforms.

- Olist Store: Connects merchants to major Brazilian marketplaces like Mercado Libre, Amazon, and Americanas as a single, unified seller storefront.
- Olist Shops: A mobile-first e-commerce builder that is natively integrated with social media platforms, allowing users to create their own standalone digital storefronts.

## ***2. Logistics (Olist Pax)***

Olist operates a cloud-based logistics and fulfillment network. It aggregates the shipping volumes of its thousands of SMB sellers to negotiate better rates with carriers, simplifying warehousing, inventory management, and last-mile delivery for small merchants.

## ***3. Financial Services (Olist Credit & Olist Pay)***

To support the operational health of small businesses, the platform offers embedded financial tools.

- Olist Credit: Provides working capital and business loans for merchants.
- Olist Pay: Facilitates integrated payment processing and financial management solutions.

## **Who Olist Serves**

Olist empowers local shopkeepers, digital brands, and wholesalers who need assistance navigating complex e-commerce integrations. Instead of spending money and time building separate systems for each channel, sellers use Olist to import catalogs from an ERP, manage stock across multiple platforms, process payments, and fulfill orders.

## **Financials**

Olist is privately owned by its founder, Tiago Dalvi, alongside a group of major venture capital and private equity investors

Latest Financial Data: May 17, 2026

Revenue: \$84.9M ARR (Annual Recurring Revenue)

Assets: Olist is a Brazilian Series E unicorn valued at \$1.5 billion

Liabilities: Not publicly disclosed

Return on Equity: Not Available in the public domain

Number of Employees: 1,000

#### Key Executives

Tiago Dalvi, 41, Founder, CEO

Patricia Tourinho, VP of HR and Legal (Chief Human Resources Officer)

Sandra Montes, CGMO (Chief Growth Marketing Officer)

Felippe Galeb, 38, CPO (Chief Product Officer)

#### Major Competitors

VTEX

Mirakl

Nuvemshop

Shopify

Lomadee

AnyMarket

### 2.3 Business and Analysis Opportunity

The growth of e-commerce has created significant opportunities for businesses to use data analytics to improve customer experience, optimize operations, and increase sales performance. This project focuses on analyzing the Olist Brazilian E-Commerce Public Dataset

available on Kaggle, which contains real-world Brazilian e-commerce transaction data collected between 2016 and 2018.

The dataset includes approximately 99,441 customers, 99,441 orders, 100,000 review records, 32,951 products, and 3,095 sellers across multiple product categories. The dataset consists of multiple relational CSV files containing business-related variables such as product category, order status, payment type, freight cost, review score, customer location, and delivery performance. After joining the selected datasets, the final analytical dataset contains more than 100,000 observations suitable for analytics, visualization, and forecasting.

The primary objective of this project is to identify the characteristics associated with high and low customer satisfaction within the Olist Brazilian E-Commerce platform. Using clustering techniques, products and orders will be grouped based on customer review scores and operational characteristics such as delivery performance, freight costs, product categories, and seller attributes. The analysis aims to identify patterns that distinguish highly rated customer experiences from poorly rated experiences.

In addition, the project will forecast future sales revenue and order demand trends using historical transaction data. Time-series forecasting techniques will be applied to identify seasonal sales patterns and estimate future business performance. These insights can support inventory planning, operational decision-making, and customer experience improvement initiatives.



Monthly sales revenue trends observed in the Olist Brazilian E-Commerce Dataset, illustrating seasonal purchasing behavior and supporting future sales forecasting analysis.

## 2.4 Research Questions and Sub-Questions

**RQ1: Can clustering identify distinct groups of products/orders based on customer satisfaction and operational characteristics?**

### RQ1 Sub-Questions

- What clusters emerge when products and orders are grouped according to customer satisfaction ratings?
- Which delivery performance metrics are associated with highly rated clusters?
- How do freight costs differ across customer satisfaction clusters?
- Which product categories are most common in highly rated clusters?
- What operational characteristics are associated with low-rated clusters?

**RQ2: How accurately can future sales trends be forecast using historical transaction data?**

### RQ2 Sub-Questions

- What seasonal patterns exist in historical sales data?

- Which forecasting model provides the highest predictive accuracy?
- How do monthly order volumes change over time?
- How can forecasting support inventory and operational planning?

## 2.5 Hypotheses and Planned Analytical Techniques

**H1: Products and orders belonging to clusters with shorter delivery times, lower freight costs, favorable product categories, and stronger operational performance will exhibit significantly higher customer review scores than other clusters.**

### *Analytical Techniques:*

- K-Means Clustering
- Cluster Profiling
- ANOVA
- Descriptive Statistics

The analysis will identify distinct customer satisfaction clusters and evaluate differences in operational and transactional characteristics across clusters.

**H2. Historical sales transaction patterns contain sufficient seasonal and trend information to accurately forecast future sales revenue and order demand.**

### *Analytical Techniques:*

- Time-Series Analysis
- Moving Averages
- Exponential Smoothing
- ARIMA

The analysis will evaluate sales seasonality and forecasting performance using historical order and payment data.

## 2.6 Dataset Source

The project will utilize the Olist Brazilian E-Commerce Dataset available publicly through Kaggle.

### Dataset Information

- Dataset Name: Olist Brazilian E-Commerce Public Dataset
- Source: Kaggle
- Time Period Covered: 2016–2018
- Country: Brazil
- Number of Orders: Approximately 100,000
- Dataset Includes:
  - Customer information
  - Orders and order items
  - Product categories
  - Payments
  - Seller information
  - Customer reviews
  - Shipping and delivery data
  - Geolocation data

## 2.7 Measurements

The project will use transactional, operational, and customer-related variables from the Olist Brazilian E-Commerce Dataset to measure customer satisfaction and forecast sales trends.

For Research Question 1, Customer satisfaction will be measured using the **review\_score** variable (1–5 scale).

Additional clustering variables may include:

- Delivery Time
- Delivery Delay
- Freight Value

- Product Price
- Product Category
- Seller Location
- Order Status
- Payment Type

Cluster quality will be evaluated using silhouette scores and cluster profiling techniques. Cluster averages and distributions will be compared to identify characteristics associated with highly rated customer experiences.

For Research Question 2, Sales forecasting will use:

- Monthly Order Volume
- Monthly Revenue
- Payment Value
- Order Counts

These variables will be aggregated into monthly time-series datasets to identify trends, seasonality, and forecast future sales performance.

Data visualizations such as line charts, bar charts, and dashboards will also be used to support analytical findings and business recommendations.

## 2.8 Planned Methodology

This project will use the Olist Brazilian E-Commerce Public Dataset obtained from Kaggle. The dataset contains transactional and customer-related information collected between 2016 and 2018, including customer details, order information, payments, product categories, reviews, and delivery performance. Multiple relational CSV files will be combined using common identifiers such as `order_id`, `customer_id`, and `product_id` to create an analytical dataset containing more than 100,000 observations.

The project methodology will begin with data collection, integration, and preprocessing. Relevant datasets including orders, customers, products, payments, and reviews will be merged and cleaned to handle missing values, duplicate records, inconsistent formats, and data quality

issues. Date variables will be transformed into monthly and yearly formats to support trend analysis and forecasting.

Exploratory Data Analysis (EDA) will be conducted to understand customer satisfaction patterns, operational performance metrics, and sales trends. Descriptive statistics and SQL aggregation techniques will be used to analyze order frequency, sales revenue, product category performance, and regional purchasing behavior.

For Research Question 1, clustering analysis will be performed using customer satisfaction and operational variables. Customer review scores will serve as the primary customer satisfaction measure, while delivery performance, freight costs, product categories, payment methods, and seller characteristics will be used to construct and profile clusters. K-Means clustering will be applied to identify distinct groups of products and orders. Cluster profiling and comparative analysis will then be used to determine the characteristics associated with highly rated and poorly rated customer experiences. ANOVA and descriptive statistics may be used to evaluate statistically significant differences between clusters.

For Research Question 2, historical transaction data will be analyzed using time-series forecasting techniques to identify monthly sales trends and future demand patterns. Monthly sales revenue and order volume will be aggregated and used to develop forecasting models for future sales estimation.

The project will utilize SQL for data querying and aggregation, Python for data preprocessing and statistical analysis, and Power BI/Tableau for dashboard development and data visualization. The final analysis will provide actionable business insights and recommendations related to customer behavior, operational efficiency, and sales forecasting in the e-commerce industry.

## 2.9 Computational Methods and Outputs

The project will use multiple computational and analytical techniques to examine customer satisfaction patterns and sales forecasting trends within the Olist Brazilian E-Commerce Dataset.

SQL will be used for data extraction, table joins, aggregation, and querying across multiple relational datasets. Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn will be utilized for data preprocessing, exploratory data analysis (EDA), statistical analysis, and forecasting. Power BI or Tableau will be used to develop interactive dashboards and visualizations for business insights.

For Research Question 1, unsupervised machine learning techniques, including K-Means clustering, will be used to identify groups of products and orders with similar customer satisfaction characteristics. Cluster evaluation metrics, cluster profiling, descriptive statistics, and ANOVA will be used to compare high-performing and low-performing clusters. Visualizations such as cluster plots, boxplots, and heatmaps will be used to interpret results.

For Research Question 2, for sales forecasting analysis, historical transaction data will be aggregated into monthly sales trends and analyzed using time-series forecasting techniques, moving averages, and predictive analytics models to estimate future sales demand and revenue trends.

The expected outputs of this project include analytical dashboards, sales trend visualizations, customer segmentation insights, forecasting models, and business recommendations that support data-driven decision-making in e-commerce operations.

## 2.10 Output Summaries

### **Output 1 – Customer Satisfaction Cluster Profiles**

The first output will identify distinct clusters of products and orders based on customer review scores and operational characteristics. Cluster profiling will reveal the attributes associated with highly rated and poorly rated customer experiences, including delivery performance, freight costs, seller characteristics, and product categories.

### **Output 2 – Customer Satisfaction Driver Analysis**

The second output will evaluate the operational and transactional factors associated with customer satisfaction. Comparative analysis across clusters will help identify the characteristics that most strongly influence customer review scores and overall customer experience.

### **Output 3 – Sales Forecasting Dashboard**

The third output will forecast future monthly sales revenue and order demand using historical transaction data. Results will identify seasonal sales patterns, future demand fluctuations, and potential revenue growth opportunities.

Additional outputs will include Power BI/Tableau dashboards, KPI summaries, forecasting visualizations, cluster analysis reports, and business recommendations designed to improve customer satisfaction and operational efficiency.

#### **2.11 Campaign Implementation**

The customer satisfaction clustering results can help businesses identify operational practices associated with positive customer experiences. Organizations can use these insights to reduce delivery delays, optimize shipping strategies, improve seller performance, and focus marketing efforts on product categories associated with higher customer satisfaction.

Cluster-specific recommendations can support customer retention strategies and service quality improvements. Sales forecasting results can assist with inventory planning, demand management, resource allocation, and seasonal marketing campaigns.

### **3. Literature Review**

#### **3.1 Introduction**

The rapid growth of e-commerce has transformed the retail industry by enabling businesses to reach customers through digital channels while generating large volumes of transactional and customer-related data. As online marketplaces continue to expand, organizations increasingly rely on data analytics to understand customer behavior, improve customer satisfaction, optimize operations, and forecast future demand. Customer satisfaction is particularly important in e-commerce because consumers can easily switch between competing platforms when expectations regarding product quality, delivery performance, and service reliability are not met. At the same time, accurate sales forecasting supports inventory management, operational planning, and strategic decision-making.

The Brazilian e-commerce market has experienced significant growth over the past decade, creating opportunities for organizations to utilize advanced analytical techniques to improve business performance. The Olist Brazilian E-Commerce Dataset provides a comprehensive source of transactional, operational, and customer review information that enables the study of customer satisfaction patterns and sales forecasting. This literature review examines prior research related to customer satisfaction in e-commerce environments, customer segmentation and clustering techniques, operational factors influencing customer experience, and sales forecasting methodologies. The review also identifies gaps in existing research that motivate the current study.

#### **3.2 Customer Satisfaction in E-Commerce**

Customer satisfaction has long been recognized as one of the most important determinants of business success in both traditional and online retail environments. Satisfaction reflects the degree to which customer expectations are met or exceeded during the purchasing process. In e-commerce settings, customer satisfaction is influenced by multiple factors including product quality, pricing, website usability, customer service, shipping performance, and post-purchase experiences.

Prior studies have consistently shown that customer satisfaction significantly influences customer loyalty, repeat purchasing behavior, and positive word-of-mouth recommendations. Satisfied customers are more likely to make future purchases and recommend online retailers to

others, whereas dissatisfied customers often share negative experiences that can damage brand reputation. Researchers have found that delivery reliability and fulfillment performance are among the strongest predictors of customer satisfaction because customers increasingly expect rapid and accurate order fulfillment.

Review and rating systems have become essential mechanisms for measuring customer satisfaction in digital marketplaces. Online review scores provide direct feedback regarding customer experiences and are frequently used as indicators of service quality. Research suggests that review ratings capture customer perceptions of product quality, delivery efficiency, communication effectiveness, and overall transaction satisfaction. Consequently, review scores are commonly used as dependent variables in customer satisfaction research.

### **3.3 Operational Drivers of Customer Satisfaction**

A substantial body of literature has examined the operational factors that influence customer satisfaction in online retail environments. Delivery performance is frequently identified as one of the most influential determinants of customer experience. Studies have shown that delayed deliveries often result in lower customer ratings, increased complaints, and reduced customer retention. Conversely, shorter delivery times and accurate fulfillment processes contribute positively to customer perceptions.

Freight and shipping costs also influence customer satisfaction. Research indicates that customers are highly sensitive to shipping expenses, particularly when purchasing lower-priced products. Excessive freight charges may negatively affect perceived value and overall satisfaction, even when product quality remains high. Consequently, businesses seek to optimize logistics networks and shipping strategies to balance operational costs and customer expectations.

Product category characteristics may further influence customer satisfaction levels. Certain categories, such as electronics and household goods, often generate different customer expectations regarding delivery speed, product quality, and return policies. Seller performance indicators, including responsiveness and order processing efficiency, have also been linked to customer satisfaction outcomes. Together, these findings suggest that customer satisfaction is driven by a combination of operational, logistical, and product-related factors.

### **3.4 Customer Segmentation and Clustering Techniques**

Customer segmentation is widely used in business analytics to identify groups of customers, products, or transactions that exhibit similar characteristics. Segmentation enables organizations to develop targeted strategies that improve customer experiences and operational efficiency. Traditional segmentation approaches often rely on demographic variables, whereas modern analytical techniques increasingly utilize behavioral and transactional data.

Clustering techniques represent one of the most common unsupervised machine learning approaches for customer segmentation. K-Means clustering, in particular, has been extensively applied in retail analytics because of its simplicity, scalability, and effectiveness in identifying naturally occurring groups within large datasets. Researchers have used clustering methods to classify customers based on purchasing behavior, spending patterns, service interactions, and satisfaction metrics.

Recent studies demonstrate that clustering can reveal meaningful differences between high-performing and low-performing customer segments. By examining cluster profiles, organizations can identify operational characteristics associated with favorable customer experiences and develop targeted improvement initiatives. Cluster analysis has also been applied to product and order data to uncover patterns that may not be evident through traditional statistical methods.

The application of clustering techniques to customer satisfaction analysis is particularly relevant in e-commerce environments where large volumes of transactional and review data are available. By combining customer review scores with operational variables such as delivery performance, freight costs, and seller characteristics, clustering approaches can provide actionable insights regarding the factors associated with customer satisfaction.

### **3.5 Sales Forecasting in E-Commerce**

Sales forecasting is a critical component of retail operations because it enables organizations to anticipate future demand, allocate resources effectively, and manage inventory levels. Accurate forecasting helps businesses reduce operational costs while improving customer service and product availability. As e-commerce generates large volumes of transaction data, organizations increasingly utilize predictive analytics to estimate future sales performance.

Time-series forecasting methods are among the most widely used approaches for sales prediction. These methods analyze historical observations collected over time to identify trends,

seasonal patterns, and cyclical fluctuations. Researchers have successfully applied moving averages, exponential smoothing techniques, and autoregressive integrated moving average (ARIMA) models to forecast retail demand and sales revenue.

Seasonality plays a particularly important role in e-commerce forecasting because consumer purchasing behavior often varies according to holidays, promotional events, and seasonal trends. Prior studies have demonstrated that forecasting models capable of capturing seasonal variations generally outperform simpler approaches that ignore temporal patterns. Understanding seasonal demand fluctuations enables organizations to prepare inventory, staffing, and logistics resources more effectively.

Recent advancements in machine learning have expanded forecasting capabilities beyond traditional statistical techniques. Algorithms such as Random Forest, Gradient Boosting, and Neural Networks have shown promise in complex forecasting environments. However, many studies continue to find that simpler forecasting methods perform competitively when data exhibit clear trends and seasonal patterns. Consequently, model selection should be guided by predictive accuracy and interpretability rather than complexity alone.

### **3.6 Research Gap**

Although substantial research has examined customer satisfaction and sales forecasting independently, fewer studies have investigated both topics within a single integrated e-commerce analytical framework. Existing customer satisfaction research frequently focuses on individual operational variables such as delivery performance or review ratings, while forecasting studies often concentrate exclusively on sales trends without considering customer experience dimensions.

Furthermore, relatively limited research has utilized the Olist Brazilian E-Commerce Dataset to simultaneously examine customer satisfaction segmentation and sales forecasting. Most prior studies employ traditional descriptive analyses or predictive models without incorporating unsupervised clustering techniques to identify distinct customer satisfaction profiles. There is also limited evidence regarding how operational characteristics such as freight costs, delivery performance, and product categories collectively influence customer satisfaction clusters.

This study addresses these gaps by applying K-Means clustering and cluster profiling techniques to identify high-performing and low-performing customer satisfaction groups while

simultaneously developing forecasting models to predict future sales revenue and order demand. The integration of customer satisfaction analytics and sales forecasting provides a more comprehensive understanding of e-commerce performance and supports data-driven business decision-making.

### **3.7 Conclusion**

The literature demonstrates that customer satisfaction and sales forecasting are critical components of successful e-commerce operations. Prior research highlights the importance of delivery performance, freight costs, product characteristics, and seller effectiveness in shaping customer experiences. Similarly, forecasting studies emphasize the value of time-series analysis for predicting future sales trends and supporting operational planning. Despite these contributions, opportunities remain for integrated analyses that combine customer satisfaction segmentation with sales forecasting using comprehensive e-commerce datasets. The present study seeks to contribute to the existing body of knowledge by leveraging the Olist Brazilian E-Commerce Dataset to examine customer satisfaction clusters and forecast future sales performance.

#### **4. Final Research Questions**

**RQ1:** Can clustering identify distinct groups of products/orders based on customer satisfaction and operational characteristics?

**RQ2:** How accurately can future sales trends be forecast using historical transaction data?

## 5. Exploratory Data Analysis

### 5.1 Introduction

The purpose of this exploratory data analysis (EDA) is to understand the structure, quality, and characteristics of the Olist Brazilian E-Commerce Dataset and identify patterns that support the two research questions of this study:

**RQ1: Can clustering identify distinct groups of products/orders based on customer satisfaction and operational characteristics?**

**RQ2: How accurately can future sales trends be forecast using historical transaction data?**

The Olist dataset consists of multiple relational tables containing information about customers, orders, products, sellers, payments, and customer reviews. Exploratory analysis was performed to assess data quality, understand variable distributions, identify outliers, investigate relationships among variables, and examine historical sales patterns.

### 5.2 Dataset Overview

The Olist dataset contains multiple interconnected datasets representing different components of the e-commerce process.

## Dataset Inventory

*Table 1. Record and variable counts for the seven Olist relational datasets.*

| Dataset     | Records | Variables |
|-------------|---------|-----------|
| Customers   | 99,441  | 5         |
| Orders      | 99,441  | 8         |
| Reviews     | 99,224  | 7         |
| Order Items | 112,650 | 7         |
| Payments    | 103,886 | 5         |
| Products    | 32,951  | 9         |
| Sellers     | 3,095   | 4         |

The larger number of records in the Order Items and Payments datasets indicates that individual orders may contain multiple products and multiple payment transactions. These relationships were considered during data integration to avoid duplication in forecasting analysis.

### 5.3 Data Quality Assessment

#### Missing-Value Analysis

A missing value assessment was conducted to evaluate data completeness across all datasets. The analysis revealed that most business-critical variables contain complete information. However, the Reviews dataset contains substantial missing values in textual review fields.

## Major Missing Values Identified

*Table 2. Variables with the highest missing-value percentages.*

| Variable                      | Missing % |
|-------------------------------|-----------|
| review_comment_title          | 88.34%    |
| review_comment_message        | 58.70%    |
| order_delivered_customer_date | 2.98%     |
| order_delivered_carrier_date  | 1.79%     |
| product_category_name         | 1.85%     |

The large percentage of missing review titles and review comments suggests that many customers provided ratings without written feedback. Since customer satisfaction in this study is measured using review scores, these text variables were excluded from further quantitative analysis.

Delivery-related fields contain small amounts of missing data due to canceled, unavailable, or undelivered orders.

## Duplicate Record Analysis

Duplicate analysis showed that no duplicate records exist within the Customers, Orders, Reviews, Order Items, Payments, Products, or Sellers datasets. This finding indicates strong data integrity and eliminates the need for duplicate removal procedures during preprocessing.

## Data-Type Validation

Date-related variables were converted from object format to datetime format to support temporal analysis and feature engineering.

The converted variables include:

- order\_purchase\_timestamp
- order\_approved\_at
- order\_delivered\_carrier\_date

- order\_delivered\_customer\_date
- order\_estimated\_delivery\_date

This conversion enabled the creation of operational performance metrics and sales forecasting variables.

## 5.4 Feature Engineering

Two delivery-related variables were engineered from the Orders dataset.

**Delivery Duration - Delivery Days** measures the number of days between the purchase date and customer delivery date.

**Delivery Delay** - Delivery Delay measures the difference between the actual delivery date and the estimated delivery date.

- Negative values indicate early deliveries.
- Positive values indicate delayed deliveries.

These variables provide valuable measures of operational efficiency and customer experience.

## 5.5 Customer Satisfaction and Operational Performance (RQ1)

### Analysis Overview

To support customer satisfaction analysis, Orders, Reviews, Payments, Order Items, Products, and Sellers datasets were merged into a single analytical dataset. The merged dataset contains approximately 119,140 observations. The increase in records compared to the original Orders dataset occurs because some orders contain multiple products and payment transactions.

## Descriptive Statistics

### Summary Statistics

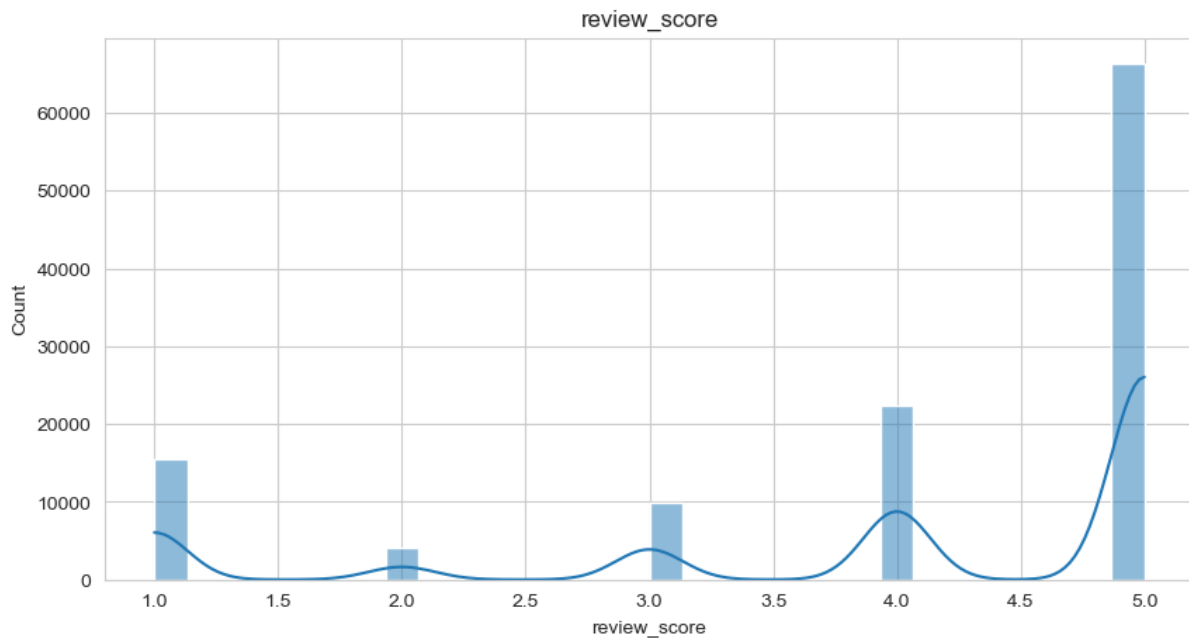
*Table 3. Summary statistics for key numerical variables in the customer satisfaction analytical dataset.*

| Variable      | Mean | Median | Maximum |
|---------------|------|--------|---------|
| Review Score  | 4.0  | 5.0    | 5       |
| Price         | 121  | 75     | 6,735   |
| Freight Value | 20   | 16     | 410     |
| Payment Value | 173  | 108    | 13,664  |
| Delivery Days | 12   | 10     | 209     |

The statistics indicate that customer satisfaction is generally high, while transaction values and delivery performance vary substantially across orders.

## Review Score Distribution

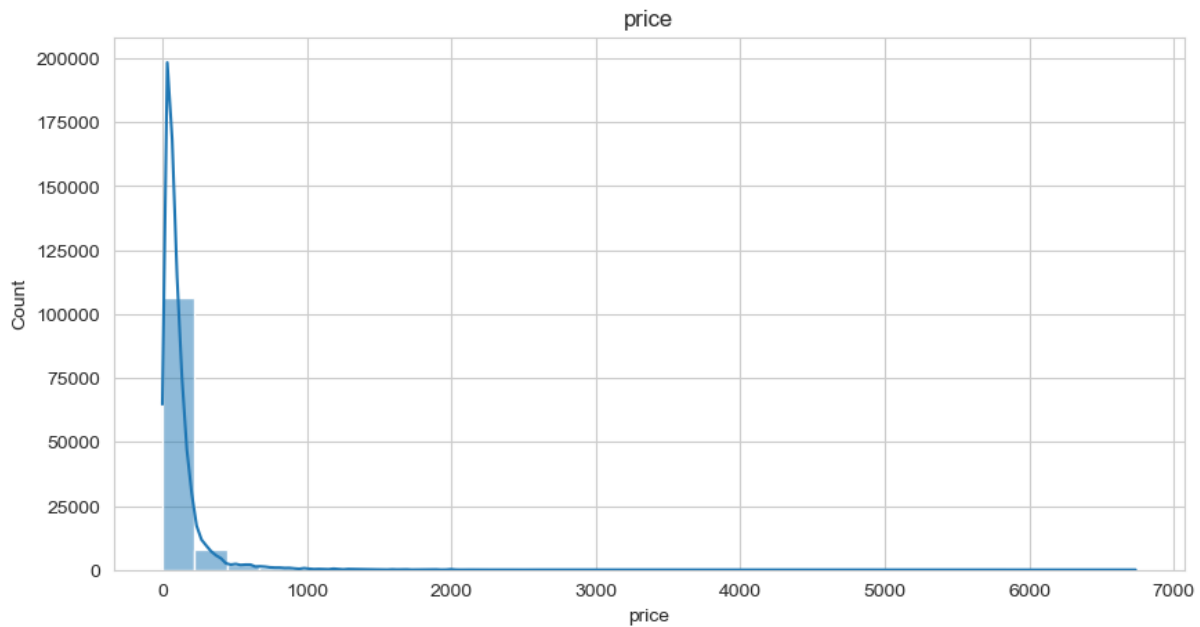
### Distribution of Customer Review Scores



The review score distribution is heavily concentrated around ratings of four and five stars, with five-star ratings dominating the dataset. This indicates that most customers reported positive purchasing experiences and overall satisfaction with the platform.

## Product Price Distribution

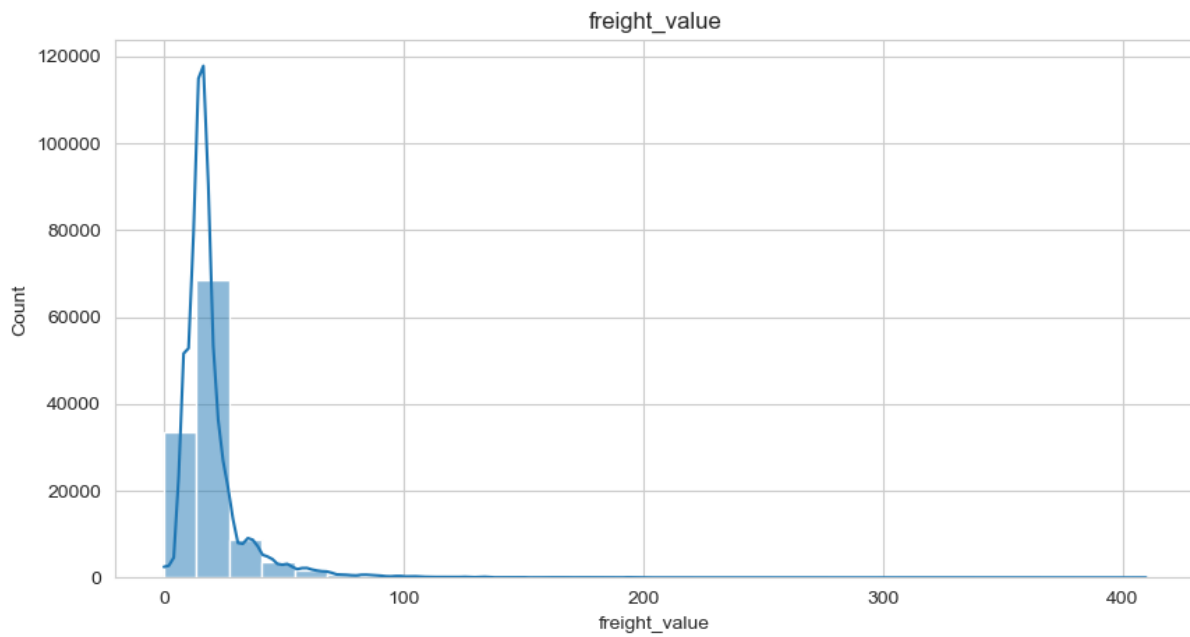
### Distribution of Product Prices



The price distribution is strongly right-skewed. Most products are sold at relatively low prices, while a small number of products account for extremely high transaction values. This pattern is typical in e-commerce environments where premium products represent a small portion of total sales.

## Freight Value Distribution

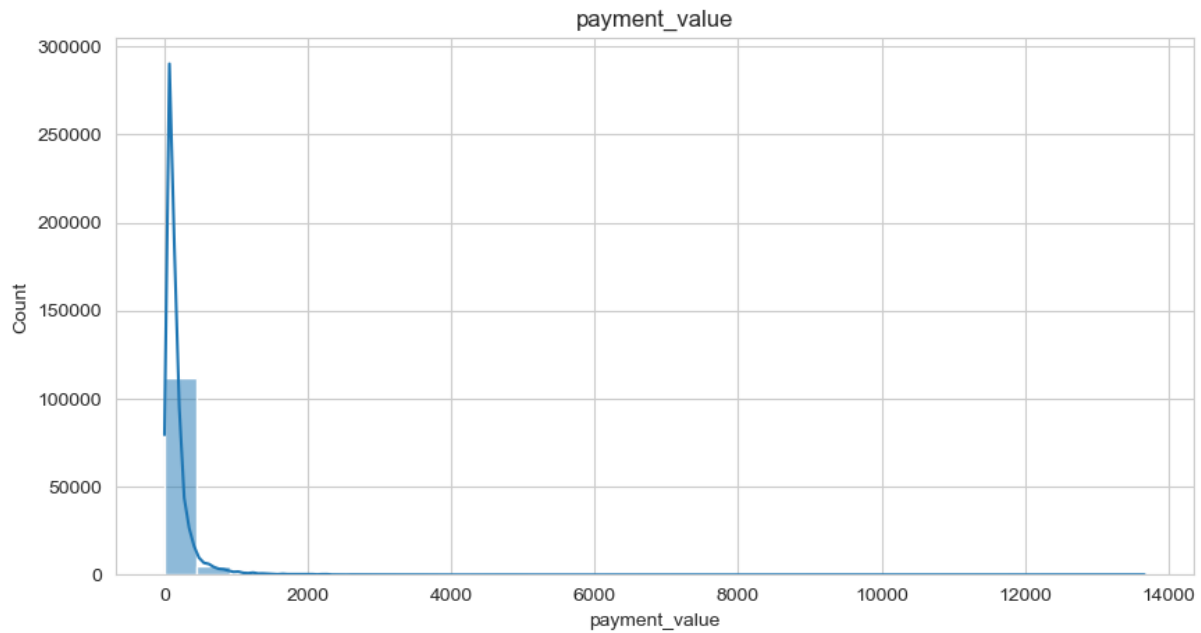
### Distribution of Freight Values



Freight values are concentrated within lower shipping cost ranges. However, several orders exhibit significantly higher freight charges, creating a long right tail in the distribution.

## Payment Value Distribution

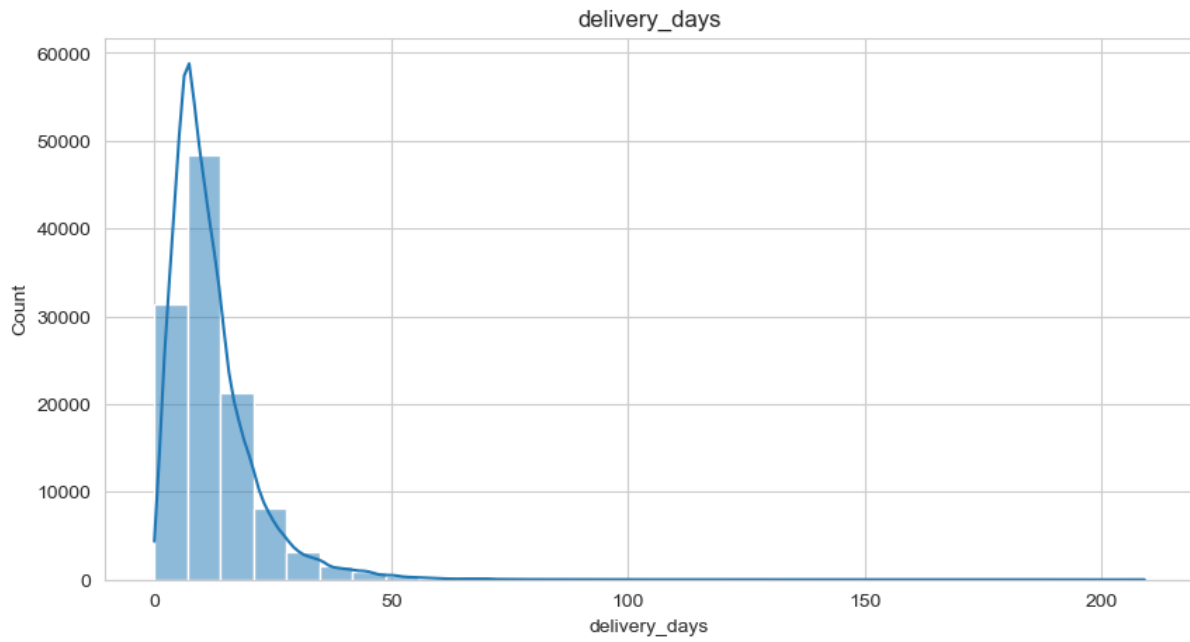
### Distribution of Payment Values



Payment values display substantial variability. Most transactions fall within lower payment ranges, while a relatively small number of transactions exceed several thousand Brazilian Reais.

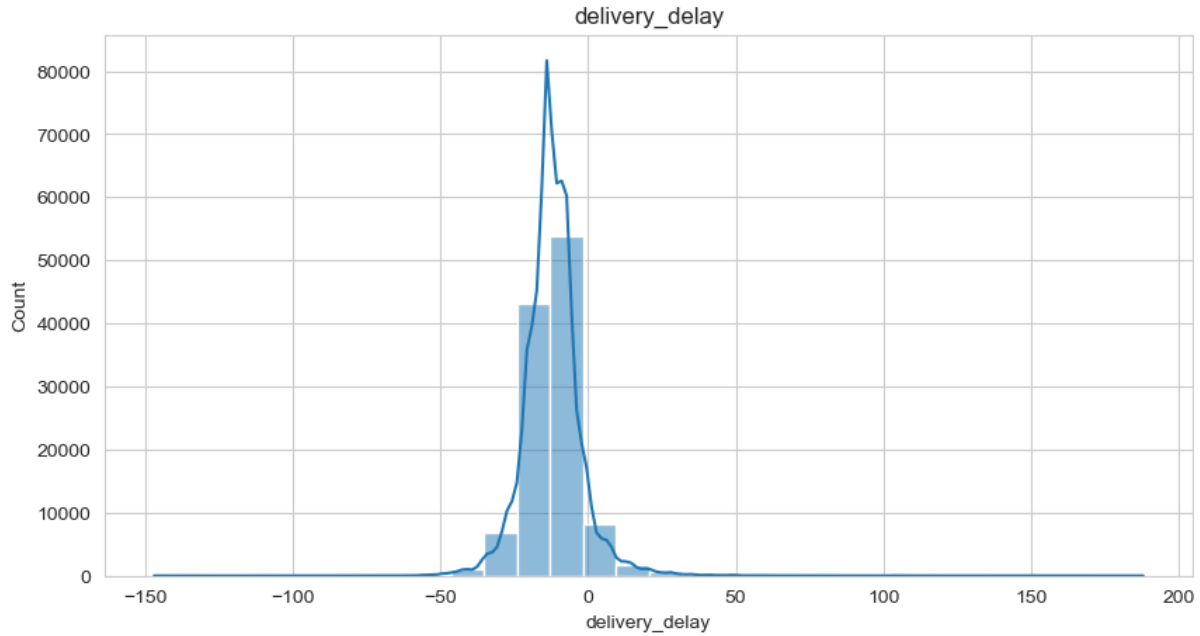
## Delivery-Performance Analysis

### Distribution of Delivery Duration



Most orders are delivered within approximately two weeks. However, a small number of orders required significantly longer delivery periods, extending beyond 100 days.

## Distribution of Delivery Delay



The delivery delay distribution is centered near zero, indicating that most deliveries occurred close to their estimated delivery dates. Negative values demonstrate that many orders arrived earlier than expected.

## Outlier Analysis

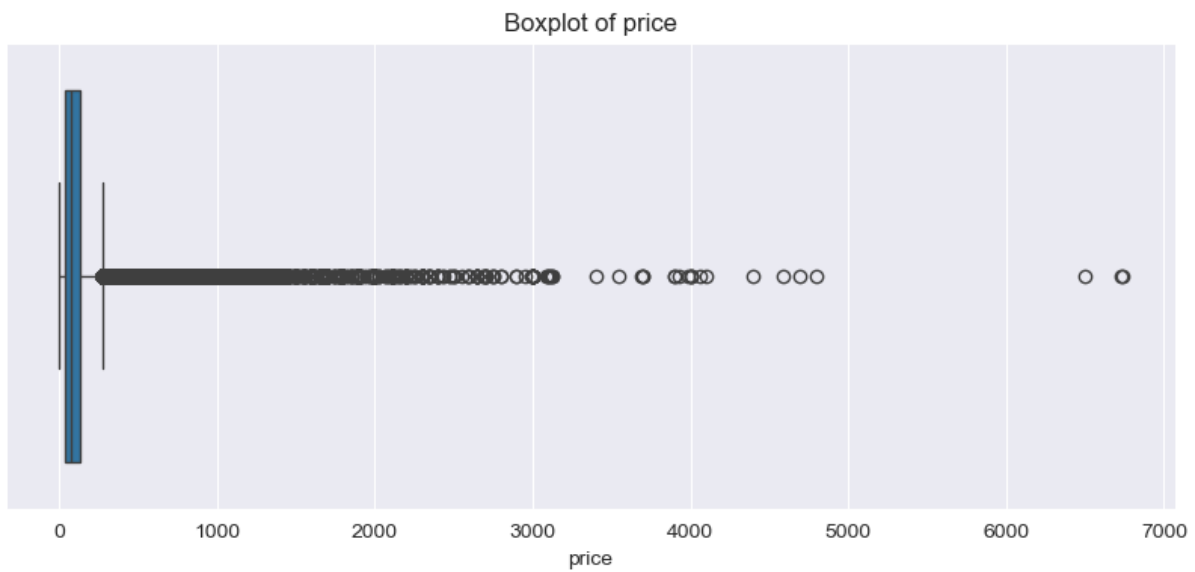
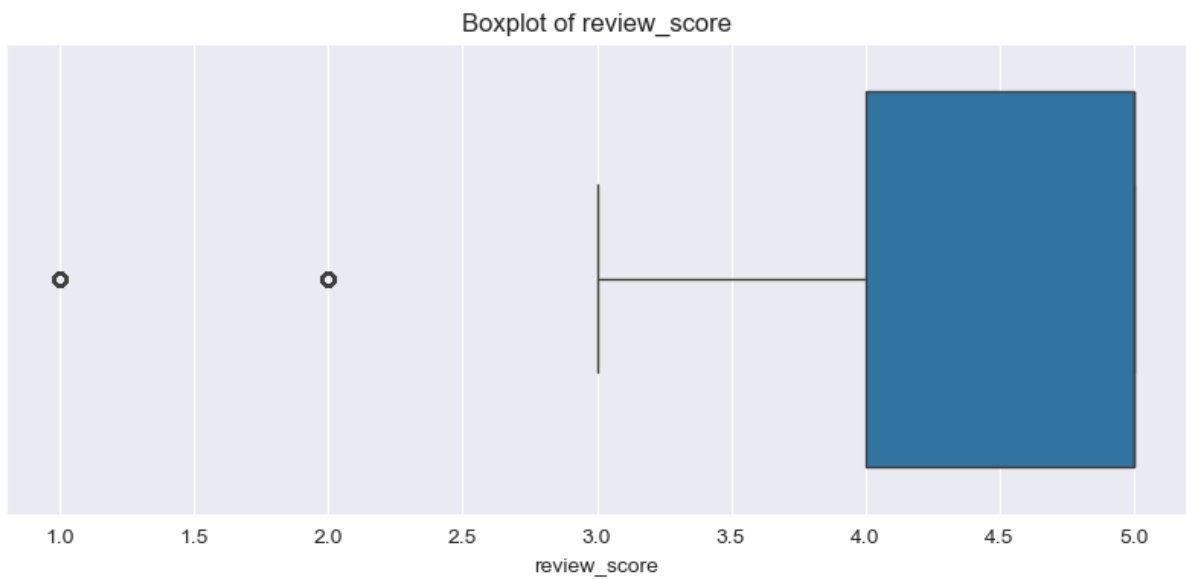
Boxplots were generated to identify unusual observations.

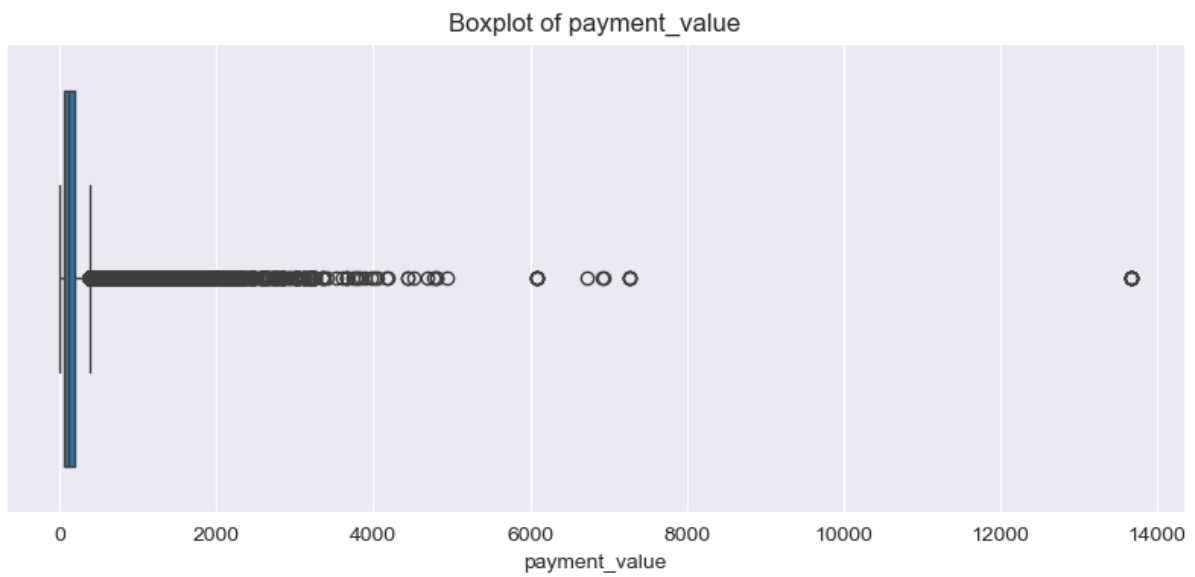
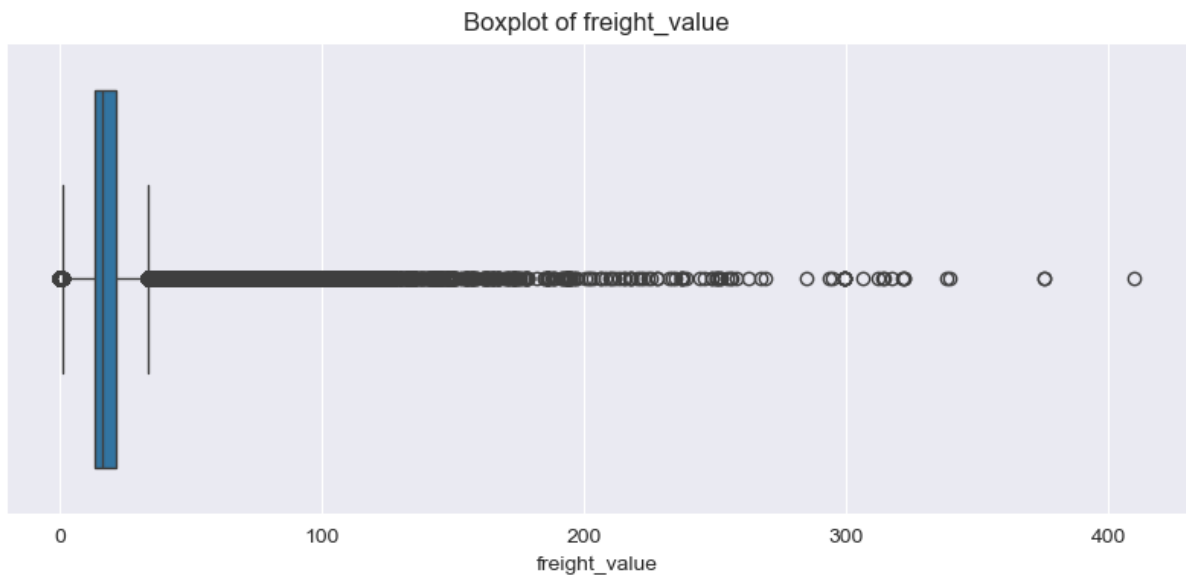
## Outlier Summary

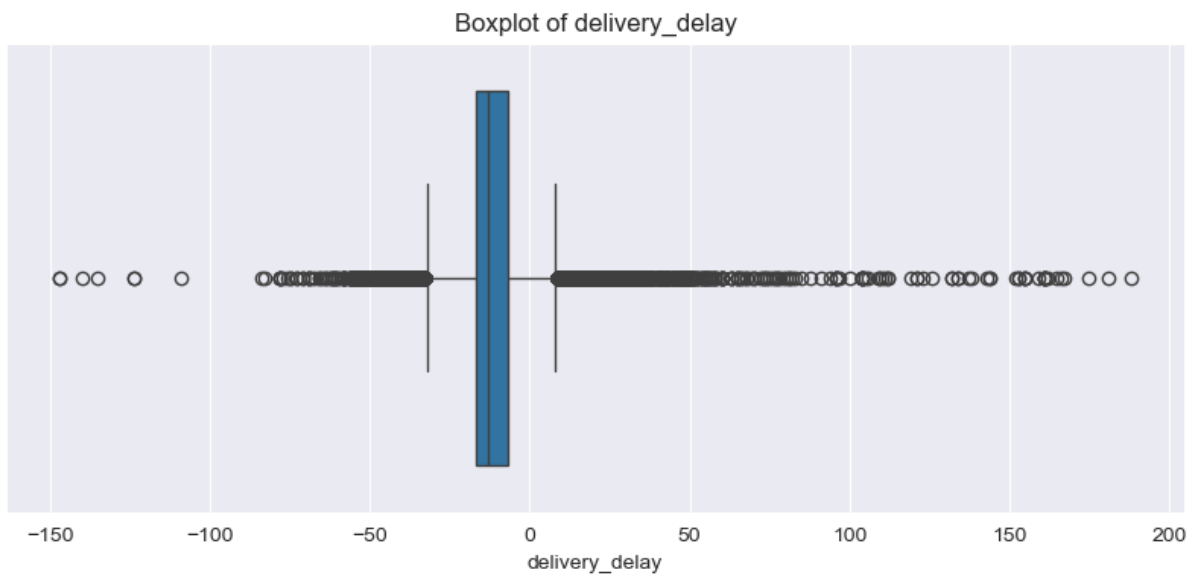
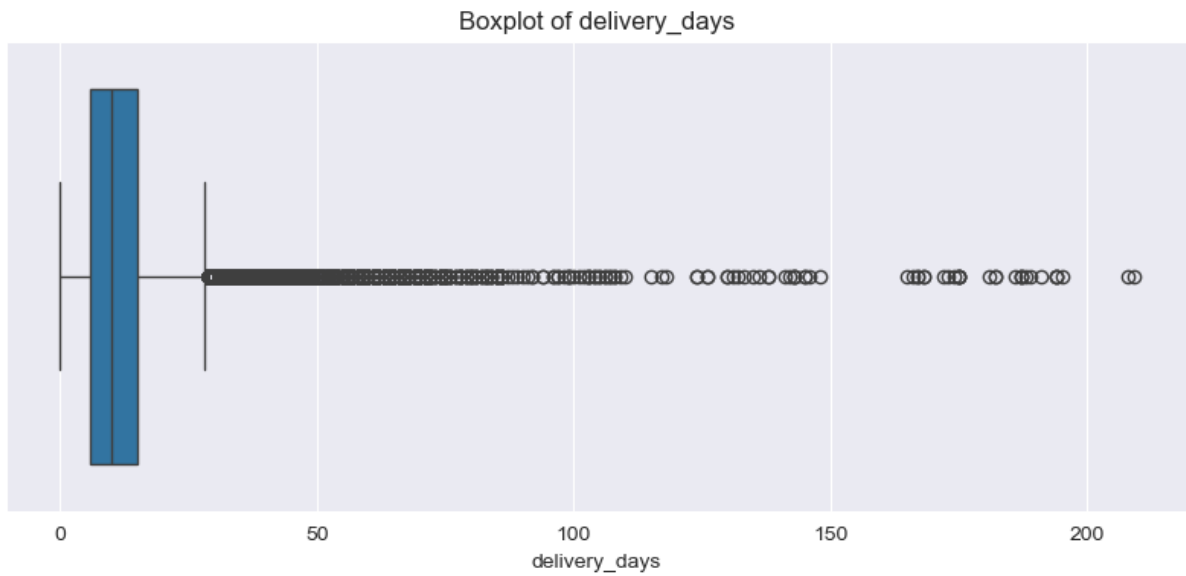
*Table 4. Outlier percentages identified by EDA boxplots, by variable.*

| Variable      | Outlier Percentage |
|---------------|--------------------|
| Freight Value | 10.69%             |
| Payment Value | 8.09%              |
| Price         | 7.44%              |
| Delivery Days | 4.88%              |

## Boxplots of Key Variables



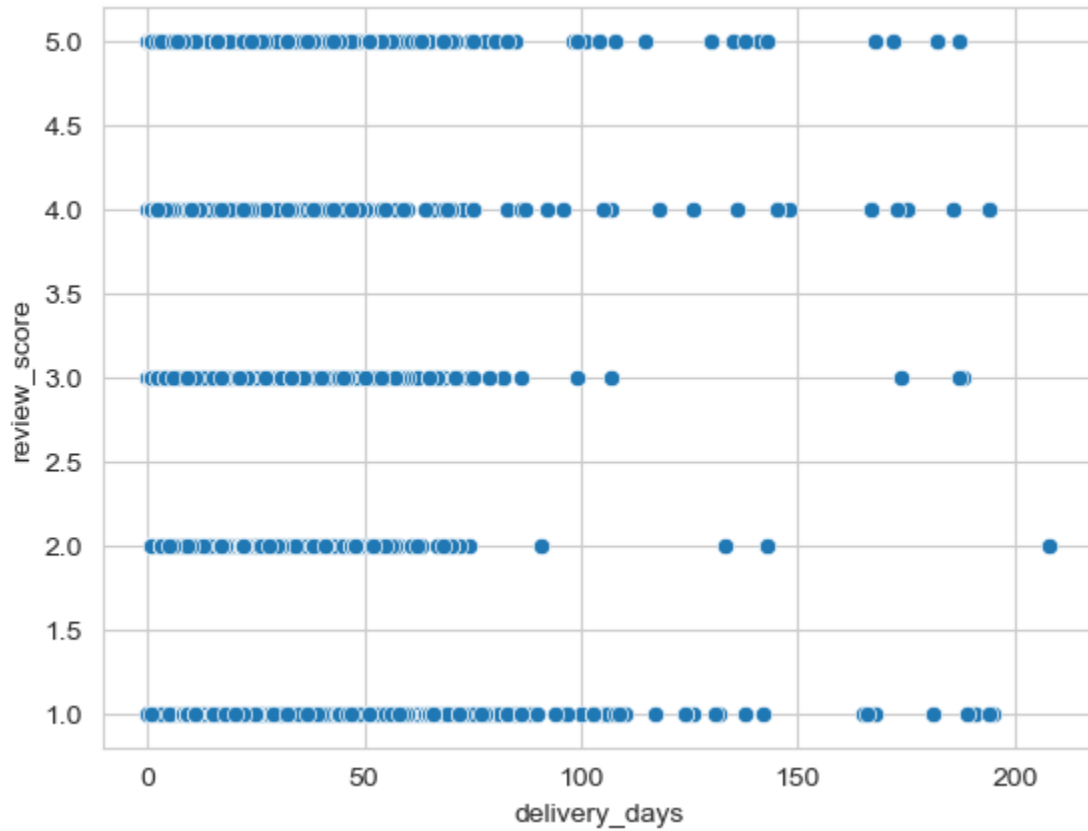




The boxplots confirm the presence of extreme observations in pricing, payment values, freight charges, and delivery times. Because these observations likely represent valid business transactions rather than errors, they were retained for subsequent analysis.

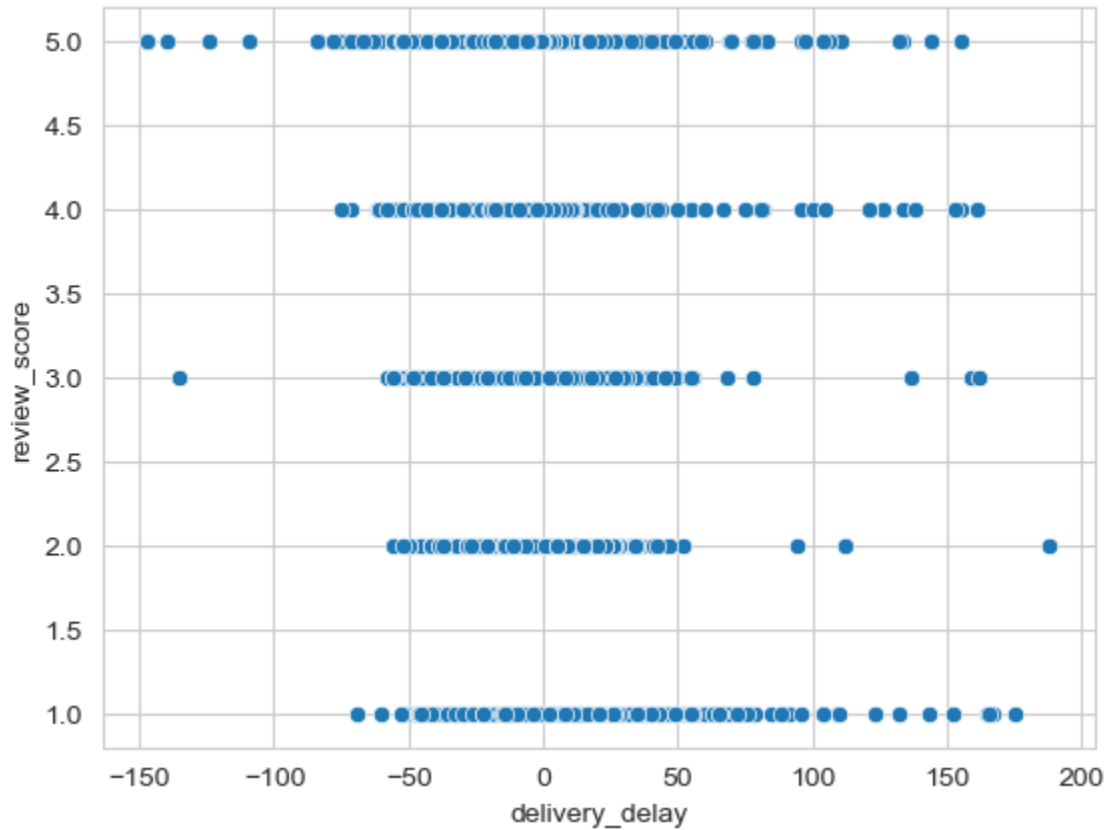
## Relationship Analysis

### Review Score vs. Delivery Duration



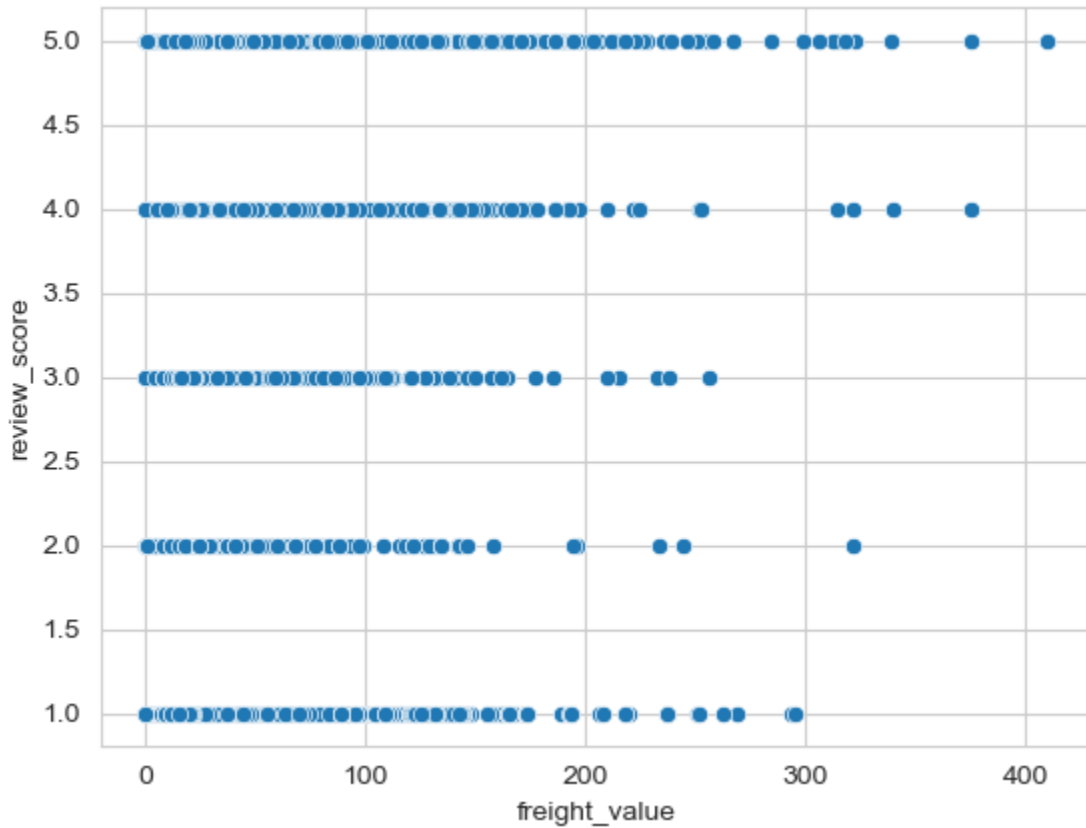
The scatterplot suggests that customers experiencing longer delivery times tend to provide lower ratings. Delivery speed appears to be an important contributor to customer satisfaction.

## Review Score vs. Delivery Delay



Orders delivered later than expected generally receive lower review scores. This finding highlights the importance of delivery reliability.

## Review Score vs. Freight Value



The relationship between freight value and review score appears weaker than delivery-related variables. However, higher shipping costs may still influence customer perceptions in certain situations.

## Correlation Analysis

### Correlation Heatmap



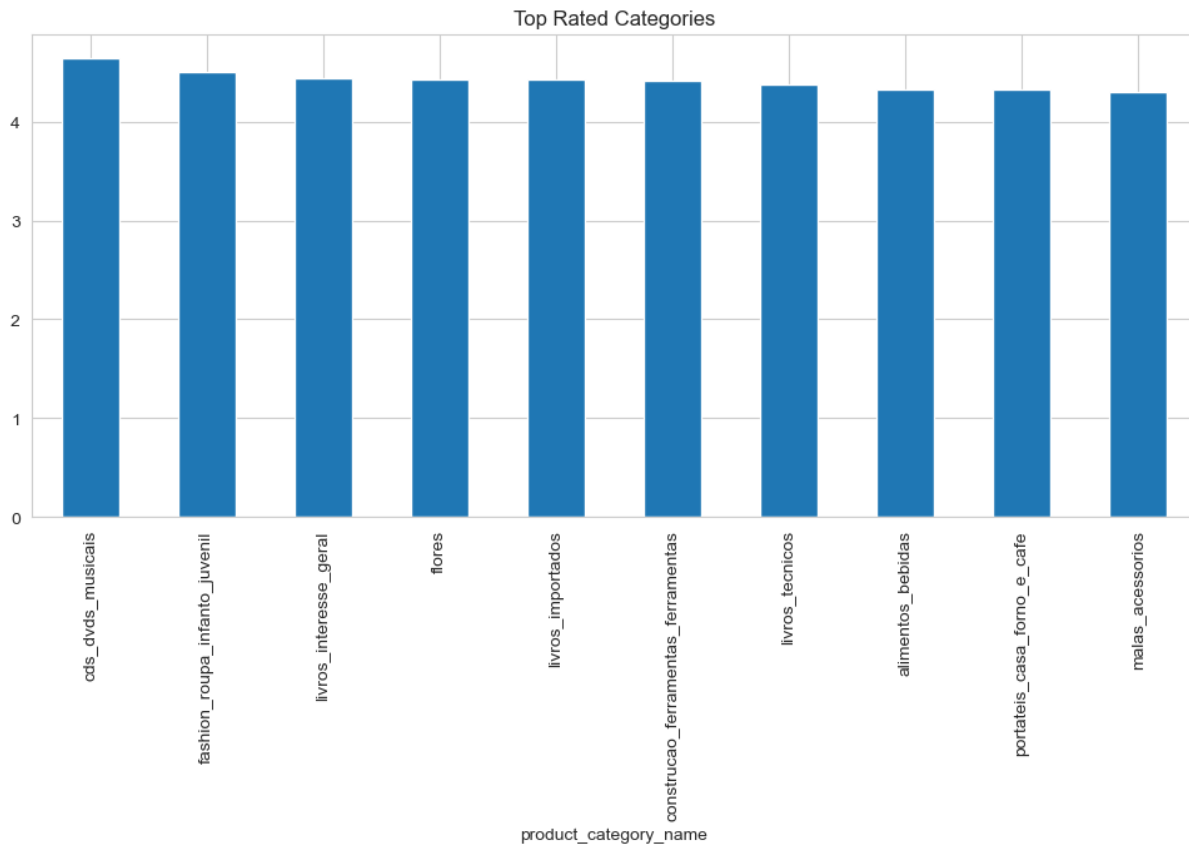
The correlation analysis reveals several meaningful relationships:

- Delivery Days and Delivery Delay: **0.6**
- Price and Payment Value: **0.76**
- Review Score and Delivery Days: **-0.30**
- Review Score and Delivery Delay: **-0.23**

The negative correlations between review scores and delivery metrics suggest that operational performance significantly influences customer satisfaction.

## Product-Category Analysis

### Top-Rated Product Categories



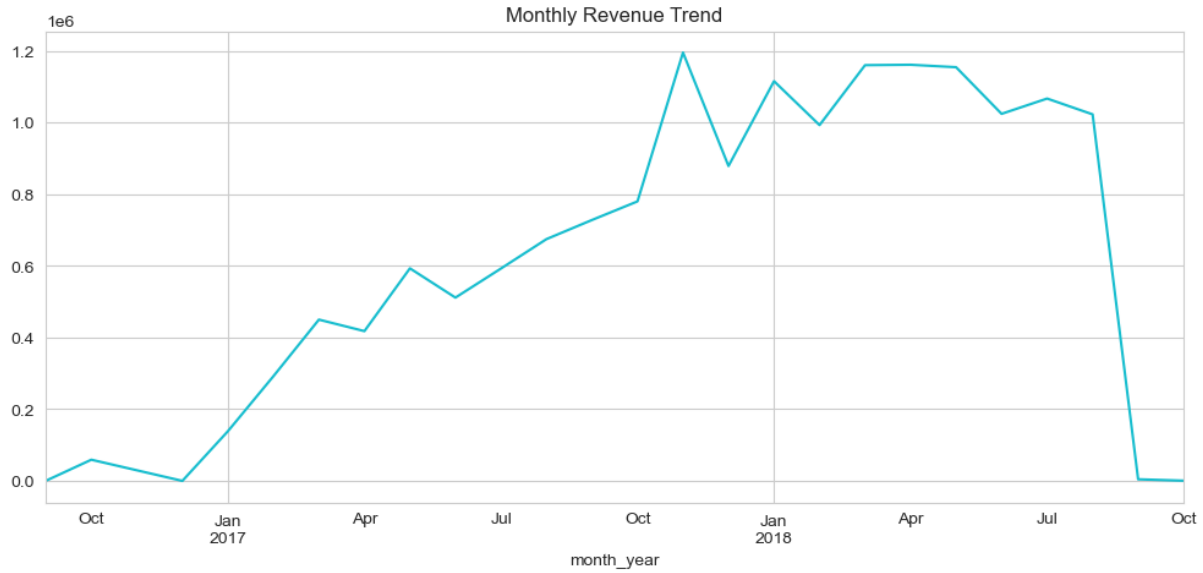
The category analysis demonstrates that certain product categories consistently achieve average review scores above 4.3 stars. These differences suggest that customer satisfaction varies across product segments and support the inclusion of product category information in clustering analysis.

## 5.6 Sales Trends and Forecasting Exploration (RQ2)

### Analysis Overview

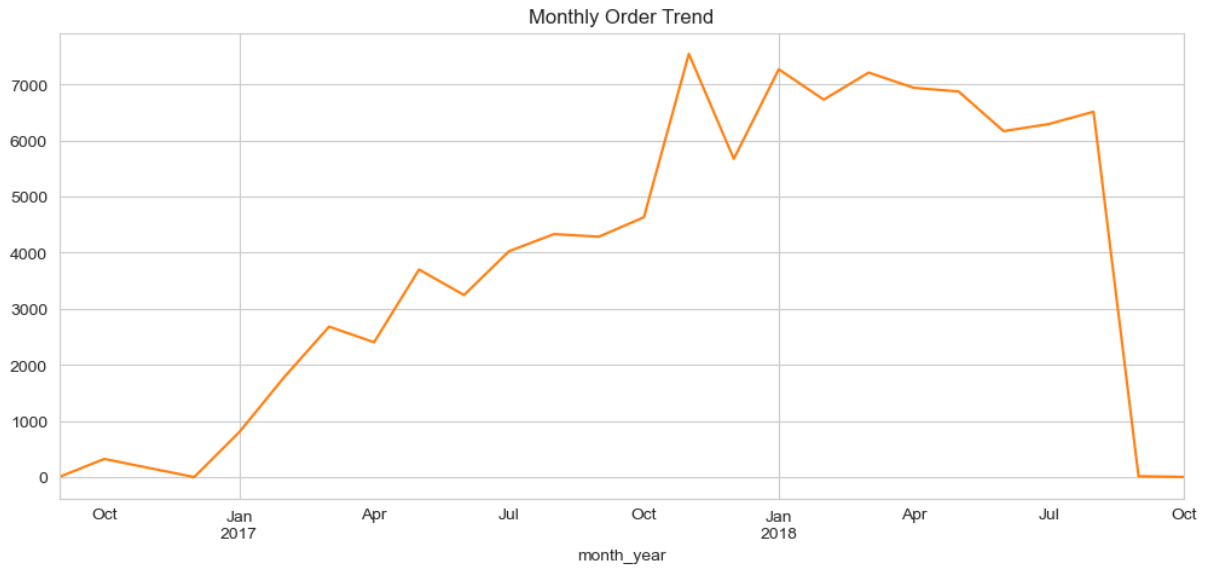
A separate forecasting dataset was created using Orders and Payments tables to avoid duplication associated with product-level records.

## Monthly Revenue Trend



Monthly revenue demonstrates a strong upward trend throughout the observation period. Revenue growth indicates increasing customer activity and marketplace expansion. Several fluctuations are observed throughout the series, suggesting the presence of seasonal purchasing behavior.

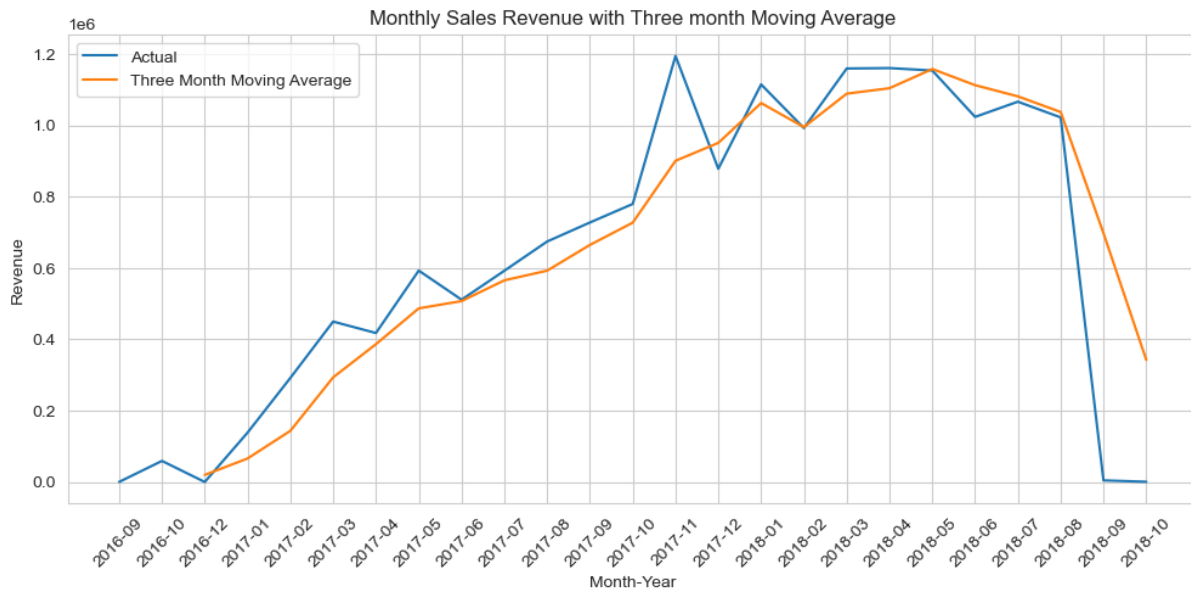
## Monthly Order-Volume Trend



Monthly order volume follows a pattern similar to revenue growth. The increase in order counts suggests that marketplace growth is driven by expanding customer participation rather than isolated high-value purchases.

## Moving-Average Analysis

### Monthly Revenue with Three-Month Moving Average



The three-month moving average smooths short-term fluctuations and clearly highlights the long-term upward trend in revenue. This confirms the existence of meaningful temporal patterns suitable for forecasting analysis.

### 5.7 EDA Findings, Limitations, and Modeling Implications

The exploratory data analysis produced several important findings.

1. The Olist dataset exhibits strong overall data quality with minimal missing values in business-critical variables.
2. Customer satisfaction is generally high, with review scores concentrated around four and five stars.
3. Delivery performance is one of the strongest operational drivers of customer satisfaction. Longer delivery times and delivery delays are associated with lower review scores.
4. Product prices, freight values, and payment values display substantial variability and contain legitimate business outliers.
5. Customer satisfaction varies across product categories, indicating meaningful differences in customer experience.
6. Monthly revenue and order volume exhibit clear growth trends over time.

7. The three-month moving average confirms an upward sales trajectory and supports the use of time-series forecasting methods.

Overall, the EDA findings provide strong support for the next phases of the project, including customer satisfaction clustering and sales forecasting model development.

## **5.8 EDA Reconciliation Note**

The EDA reflects the preliminary analytical stage originally submitted by the team. The completed analysis later used payment aggregation, review deduplication, and a revised order-item-level merge, producing 113,425 records representing 99,441 unique orders. The final notebook also identified repeated order IDs in the Payments and Reviews tables, which were resolved through this aggregation and deduplication process; this differs from the duplicate-row check performed during the EDA. The completed analysis capped the six clustering variables at their 99th percentiles before standardization, whereas the EDA retained valid outliers for exploratory purposes. The completed analysis also reported different correlation estimates because it used a different prepared sample. Final conclusions in this report are based on the completed results presented in Section 7.

## 6. Methodology

6.1 RQ1: Can clustering identify distinct groups of products/orders based on customer satisfaction and operational characteristics?

### **Data Preparation**

To investigate customer satisfaction patterns, multiple datasets from the Olist Brazilian E-Commerce Dataset were integrated. The Orders, Reviews, Payments, Order Items, Products, Sellers, and Customers datasets were merged using the common identifiers `order_id`, `product_id`, `seller_id`, and `customer_id`. This integration produced a customer satisfaction analytical dataset containing approximately 119,140 observations.

Prior to analysis, missing values, duplicate records, and variable data types were examined. Variables containing excessive missing information, such as `review_comment_title` and `review_comment_message`, were excluded because the primary customer satisfaction measure for this study is `review_score`. Delivery-related date variables were converted from object format to datetime format to facilitate feature engineering and operational performance analysis.

Two operational variables were created:

- $\text{Delivery Days} = \text{Order Delivered Date} - \text{Purchase Date}$
- $\text{Delivery Delay} = \text{Order Delivered Date} - \text{Estimated Delivery Date}$

Delivery Days measures the total waiting time experienced by customers, while Delivery Delay captures whether orders arrived earlier or later than expected. These variables were identified during exploratory analysis as important indicators of customer satisfaction.

### **Variable Selection**

Based on the exploratory data analysis, the following variables were selected for clustering:

**Table 5. Variables originally proposed for the clustering analysis. The completed K-Means model used six of these (review score, delivery duration, delivery delay, price, freight value, and payment value).**

| Variable              | Purpose                         |
|-----------------------|---------------------------------|
| review_score          | Customer satisfaction indicator |
| delivery_days         | Delivery performance            |
| delivery_delay        | Delivery reliability            |
| price                 | Product value                   |
| freight_value         | Shipping cost                   |
| payment_value         | Transaction value               |
| product_category_name | Product characteristics         |
| seller_state          | Seller location                 |
| order_status          | Order fulfillment status        |
| payment_type          | Customer payment behavior       |

The EDA revealed negative relationships between review scores and delivery performance measures, indicating that operational efficiency significantly influences customer satisfaction. Product categories also demonstrated meaningful variation in average customer ratings, supporting their inclusion within the clustering analysis.

### **Data Transformation**

Before clustering, categorical variables such as product category, seller state, order status, and payment type will be transformed into numerical representations using one-hot encoding. Numerical variables will be standardized using z-score normalization to eliminate differences in scale among variables. Standardization is required because clustering algorithms are sensitive to variable magnitude and distance calculations. The standardized variables will form the final clustering dataset.

## **Clustering Technique**

The primary analytical technique for Research Question 1 will be K-Means Clustering. K-Means partitions observations into groups by minimizing within-cluster variation while maximizing differences between clusters. The objective is to identify distinct customer satisfaction segments based on operational performance and transaction characteristics.

The optimal number of clusters will be determined using:

- Elbow Method
- Silhouette Score Analysis

The Elbow Method identifies the point at which additional clusters provide diminishing improvements in cluster cohesion. Silhouette Scores evaluate the quality of cluster separation by measuring the similarity of observations within clusters compared to observations in other clusters.

## **Cluster Profiling**

After cluster formation, descriptive statistics will be generated for each cluster.

Cluster profiles will be examined using:

- Average Review Score
- Average Delivery Days
- Average Delivery Delay
- Average Freight Value
- Average Transaction Value
- Product Category Composition
- Seller Location Distribution

This analysis will provide practical interpretations of each cluster and identify the operational characteristics associated with higher and lower customer satisfaction levels.

## **Statistical Validation**

To determine whether clusters differ significantly, Analysis of Variance (ANOVA) will be conducted on key numerical variables. ANOVA will test whether variables such as review score, delivery days, freight value, and payment value differ significantly across clusters. Statistical significance will indicate that the clustering solution captures meaningful variation within the data.

## **Evaluation Metrics**

The clustering solution will be evaluated using:

- Silhouette Score
- Within-Cluster Sum of Squares (WCSS)
- Cluster Interpretability

A higher Silhouette Score indicates stronger separation between clusters and improved clustering quality.

## **Completed RQ1 Implementation**

The completed K-Means analysis used six numerical variables: review score, delivery duration, delivery delay, item price, freight value, and payment value. Values were capped at the 99th percentile and standardized before modeling. Although categorical variables were proposed in the original methodology, they were not included in the final six-variable clustering model.  $K = 4$  was selected as the final solution based on interpretability, balance, stability, and the small silhouette-score difference from  $K = 6$ .

6.2 RQ2: How accurately can future sales trends be forecast using historical transaction data?

## **Data Preparation**

To support forecasting analysis, a separate analytical dataset was constructed using the Orders and Payments datasets. This approach avoids duplication issues associated with product-level records and ensures that revenue calculations accurately represent transaction activity. Order purchase dates were converted into datetime format and aggregated into monthly periods.

Monthly revenue and monthly order volume were calculated to create a time-series dataset suitable for forecasting.

The forecasting dataset spans the entire available transaction history from the Olist platform.

### Variable Selection

The forecasting analysis will utilize the following variables:

*Table 6. Variables used in the sales forecasting analysis.*

| Variable                 | Purpose                          |
|--------------------------|----------------------------------|
| order_purchase_timestamp | Time dimension                   |
| payment_value            | Monthly revenue calculation      |
| order_id                 | Monthly order volume calculation |

During exploratory analysis, monthly revenue and monthly order volume demonstrated a clear upward trend and evidence of temporal variation, indicating suitability for forecasting analysis.

### Time-Series Aggregation

The transactional data will be aggregated into monthly observations.

Two primary forecasting series will be created:

1. Monthly Revenue
2. Monthly Order Volume

Monthly aggregation reduces daily noise and provides a stable representation of long-term business performance.

### Forecasting Techniques

Several forecasting techniques will be evaluated.

### ***Moving Average***

A three-month moving average will be used to smooth short-term fluctuations and identify underlying sales trends.

### ***Exponential Smoothing***

Exponential Smoothing assigns greater weight to recent observations and is effective when recent sales activity is expected to influence future performance.

### ***ARIMA***

AutoRegressive Integrated Moving Average (ARIMA) models will be used to capture temporal dependencies within the sales data.

ARIMA combines:

- Autoregressive components
- Differencing operations
- Moving average components

to model trends and temporal structure within the series.

### **Model Training and Validation**

The forecasting dataset will be divided chronologically into training and testing periods. The training period will be used to fit forecasting models, while the testing period will evaluate predictive performance on unseen observations. This approach reflects real-world forecasting conditions where future values are predicted using historical data only.

### **Evaluation Metrics**

Forecasting models will be evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

Lower values indicate improved forecasting accuracy.

## **Output and Interpretation**

The final forecasting model will generate future revenue projections and expected order volumes. Forecast results will be visualized using line charts comparing actual and predicted values. The forecasts will provide insights into expected business growth patterns and support strategic decision-making regarding inventory planning, seller management, and customer demand forecasting.

## **Completed RQ2 Implementation**

The completed forecasting analysis used monthly revenue from delivered orders and evaluated five forecasting approaches: Naïve Forecasting, Three-Month Moving Average, Simple Exponential Smoothing, Holt's Linear Trend, and ARIMA (0,1,1). The monthly revenue series consisted of 21 observations, which were divided into a training period of 16 months and a holdout test period of 5 months.

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Among the models tested, ARIMA (0,1,1) achieved the best predictive forecasting performance, producing the lowest RMSE (R\$114,881) and MAPE (9.2%). Compared with the naïve forecasting baseline, ARIMA reduced RMSE by approximately 11%, demonstrating its ability to capture temporal patterns in historical sales revenue more effectively than the alternative methods.

## **6.3 Summary of the Submitted Methodological Approach**

This study utilizes two complementary analytical approaches. The first employs K-Means clustering to identify customer satisfaction segments based on operational and transactional characteristics. The second applies time-series forecasting techniques to predict future sales performance using historical revenue and order activity.

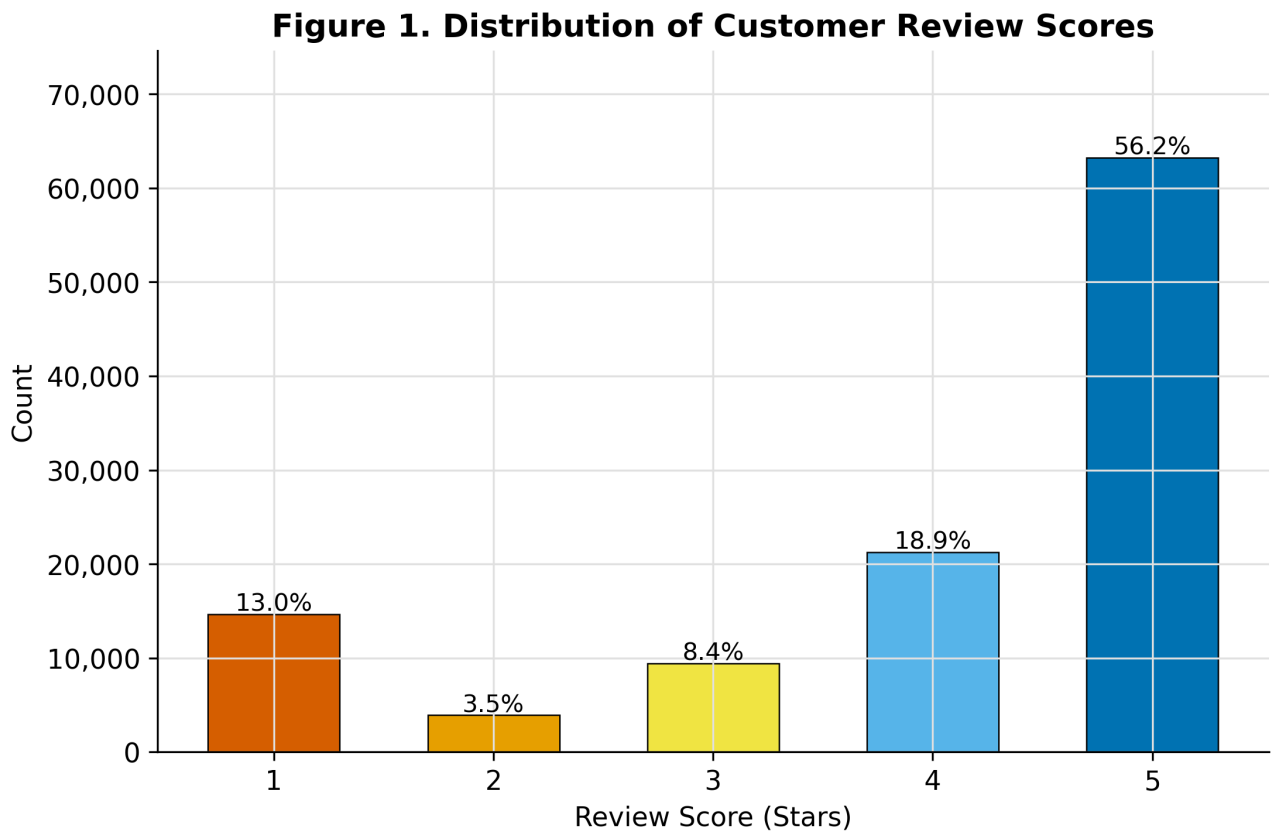
Together, these methods provide a comprehensive understanding of customer experience drivers and future business performance within the Olist e-commerce platform.

## 7. Data Visualizations and Analysis

This section presents thirteen visualizations addressing two research questions: **RQ1** — Can clustering identify distinct groups based on customer satisfaction and operational characteristics? **RQ2** — How accurately can future sales trends be forecast using historical transaction data?

**Dataset.** Seven Olist relational tables were used in this analysis: Orders, Order Items, Order Reviews, Order Payments, Products, Sellers, and the Product Category Name Translation file. The analytical dataset was constructed by aggregating payments at the order level, deduplicating reviews to one per order, and merging through left joins. The final analytical dataset contains 113,425 order-item level records representing 99,441 unique orders, consistent with the completed analytical implementation described in Section 6. The forecasting dataset uses 21 monthly aggregates of delivered orders.

**Color palette.** All visualizations use the Wong colorblind-safe palette, ensuring accessibility for individuals with deuteranopia, protanopia, and tritanopia (Wong, 2011). Colors



are assigned by meaning: blue for positive outcomes and actual values; vermilion for negative or

delayed outcomes; green for delivery-delay variables and exponential smoothing forecasts; purple for Holt’s method.

## 7.1 Customer Satisfaction and Operational Performance

### Review Score Distribution

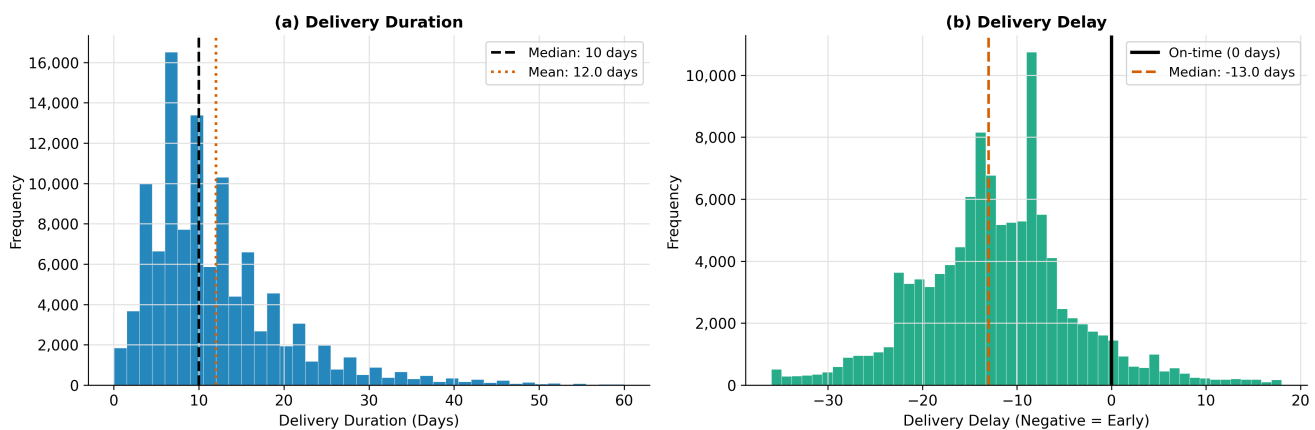
**Figure 1. Distribution of customer review scores. Colors range from warm (low) to cool (high). Percentages shown above each bar.**

Review scores are discrete, ordinal categories, so the pattern is described using category frequencies rather than skewness. Five-star ratings form the largest category, with a mean review score of 4.02 and a median of 5. A smaller concentration of one-star ratings identifies a distinct minority of dissatisfied customers. This contrast between predominantly positive ratings and a smaller dissatisfied group supports the clustering analysis.

*Limitation: Multi-item orders repeat the same review score across rows, inflating absolute counts but not the distributional shape.*

### Delivery Performance

**Figure 2. Delivery Performance Distributions**



**Figure 2. (a) Delivery duration distribution (0–60 days). (b) Delivery delay distribution (1st–99th percentile). Reference lines mark median and mean/on-time values.**

Most orders are delivered within two weeks (panel a). Approximately 92.1% of orders arrived before the estimated date, 1.3% arrived on the estimated date, and 6.6% arrived after it

(panel b). The right tail of late deliveries corresponds to lower satisfaction, as Figure 3 demonstrates.

## Review Score by Delivery Performance



**Figure 3. Mean review score by (a) delivery duration and (b) delivery delay, in 5-day bins with 95% confidence intervals. Pearson  $r$  from individual records: (a)  $r = -0.30$ ; (b)  $r = -0.23$ .**

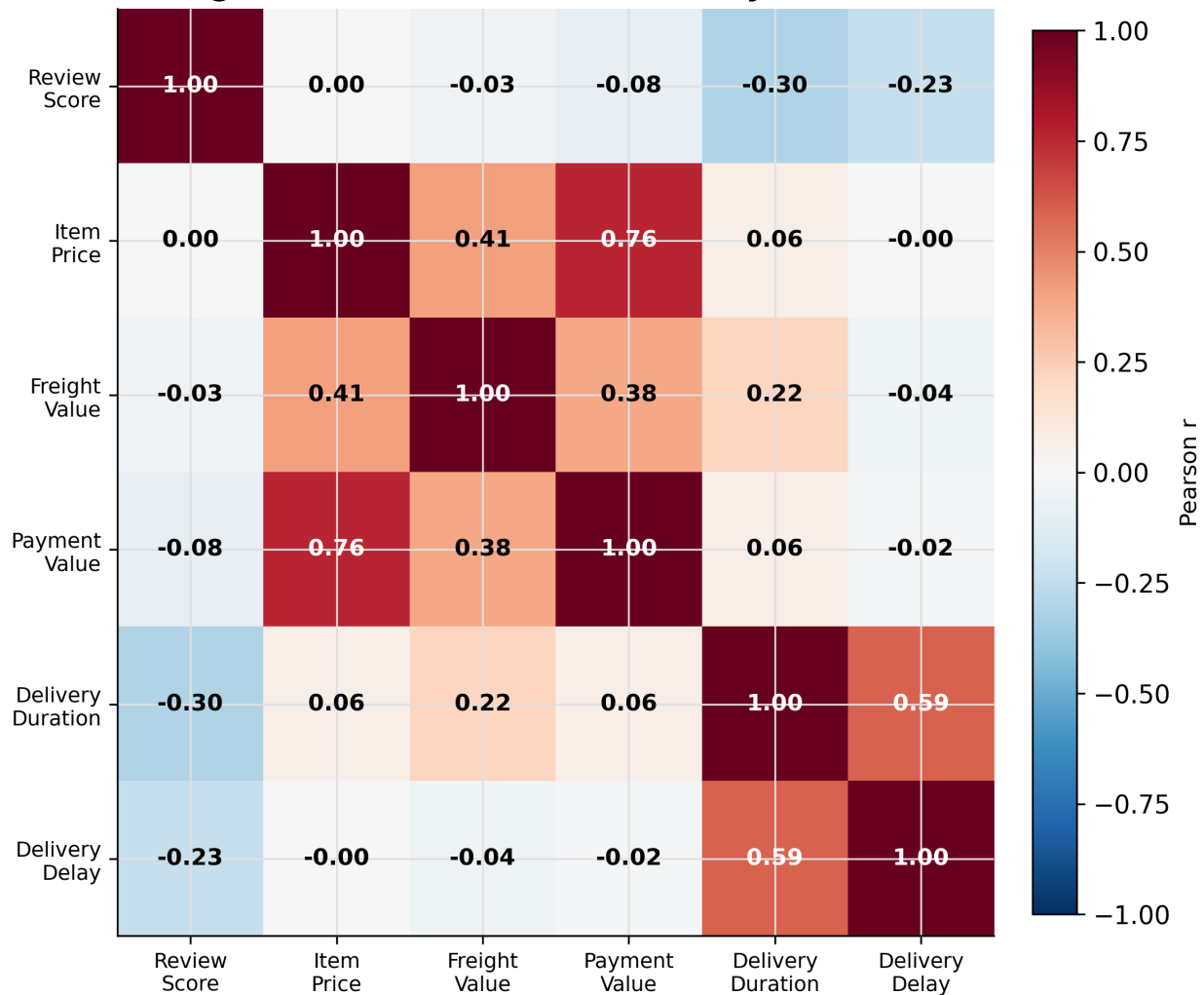
Mean review scores decline as delivery duration increases ( $r = -0.30$ ) and as delay grows more positive ( $r = -0.23$ ). These are modest but consistent associations. Confidence intervals widen at extreme bins with fewer observations. These relationships should not be interpreted causally; confounding by product type, geography, or seller quality is plausible.

## Correlation Matrix

**Figure 4. Pearson correlation matrix computed from individual records. Diverging color scale: blue (positive) to red (negative), fixed  $-1$  to  $+1$ .**

The strongest positive correlation is delivery duration with delivery delay ( $r = 0.59$ ). Review score shows negative associations with delivery duration ( $-0.30$ ) and delivery delay ( $-0.23$ ), consistent with Figure 3. These modest magnitudes suggest delivery performance explains only a small fraction of satisfaction variance.

**Figure 4. Correlation Matrix of Key Variables**



## Product Category Review Scores

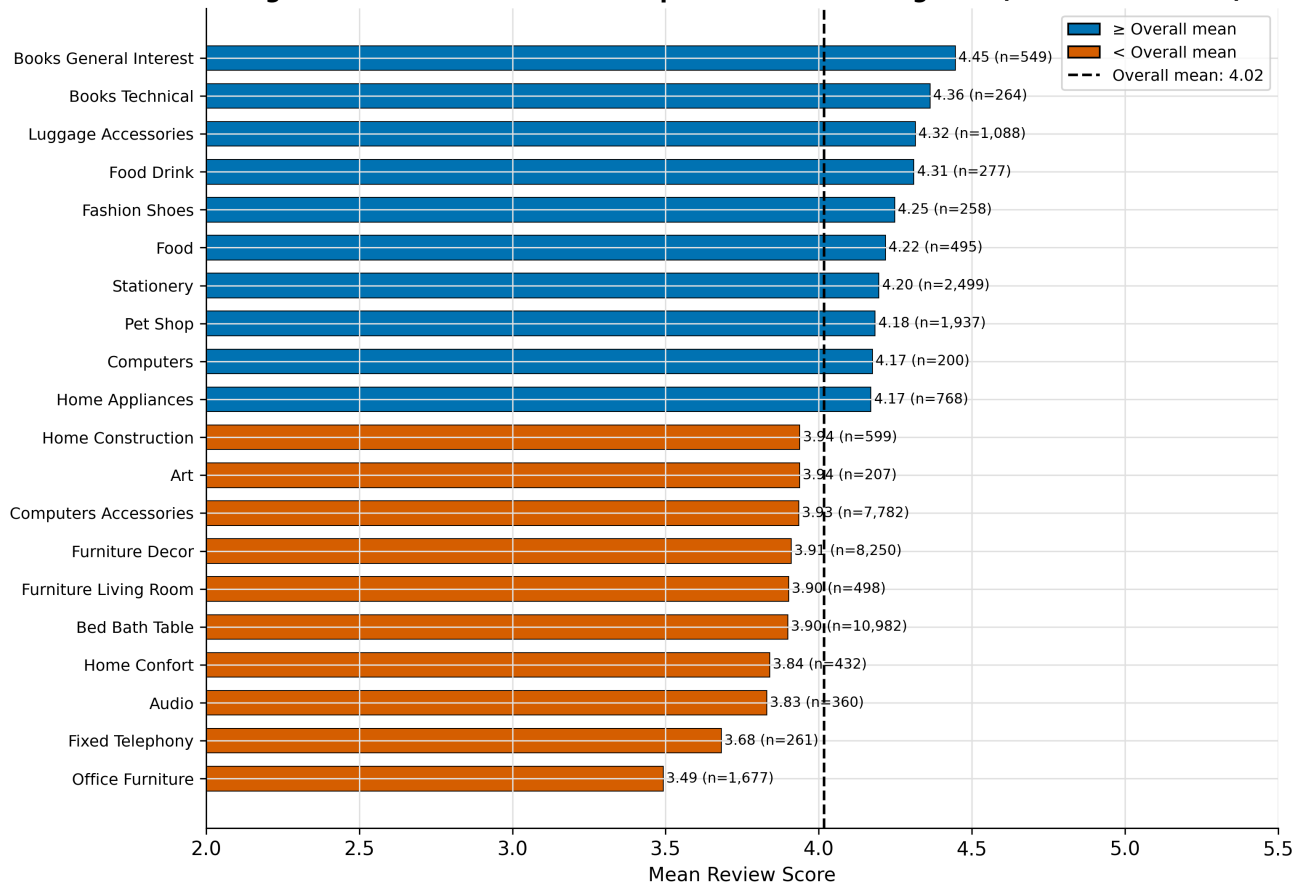
*Figure 5. Mean review score for top and bottom 10 categories (min. 200 records). Blue = at or above overall mean (4.02); vermillion = below. Sample sizes shown.*

Both the highest and lowest rated categories are displayed to show the full range of variation. The spread across categories is relatively narrow, suggesting product category is a modest differentiator of satisfaction. Sample sizes are shown beside each bar to indicate the precision of each estimate.

## 7.2 K-Means Cluster Selection and Results

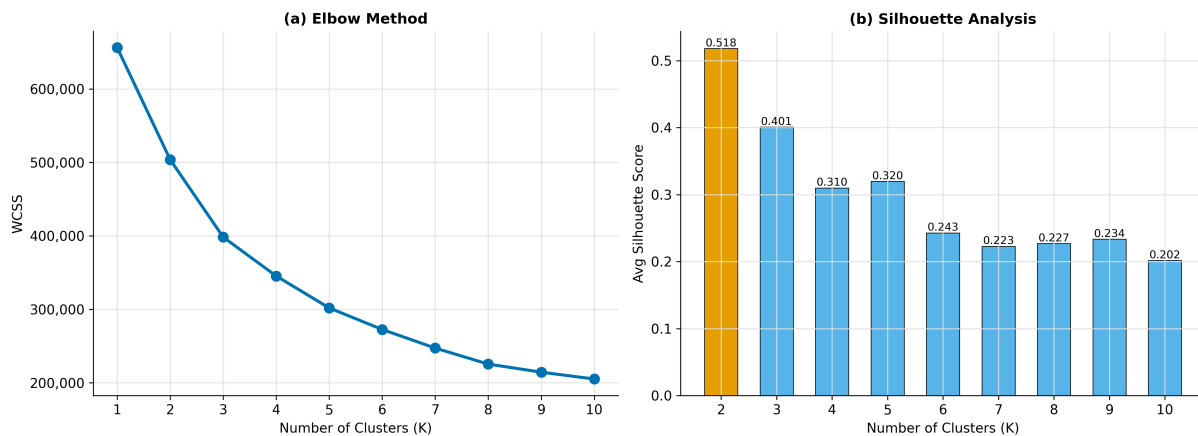
K-Means was fitted on six standardized variables: review score, delivery duration, delivery delay, item price, freight value, and payment value. Values were capped at the 99th

**Figure 5. Mean Review Score: Top & Bottom 10 Categories (Min. 200 Records)**



percentile and z-score standardized. The clustering dataset contains 109,366 complete records after removing rows with missing values in any feature.

**Figure 6. Cluster Number Selection Diagnostics**



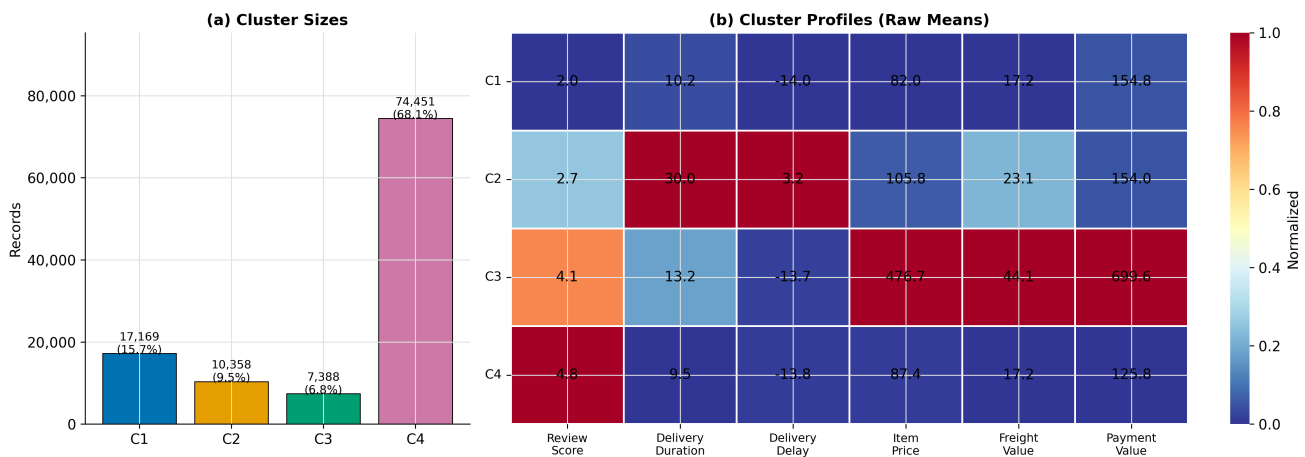
## Cluster Number Selection

*Figure 6. (a) Elbow method: WCSS by K. (b) Average silhouette score by K; orange bar marks the highest score.*

The elbow plot shows diminishing reductions in within-cluster sum of squares (WCSS) beyond  $K = 3$ – $4$ , indicating that additional clusters provide only marginal improvements in model fit.  $K = 4$  produced the highest silhouette score (**0.310**), compared with **0.243** for  $K = 6$ , indicating better cluster separation. Together with its balanced cluster sizes and strong business interpretability, the four-cluster solution was selected as the final model. Stability analysis across five random seeds further confirmed that the  $K = 4$  solution was highly consistent (standard deviation < 0.001).

**Final selection:  $K = 4$ .** The four-cluster solution produced larger, more balanced, and more interpretable groups that aligned closely with the project objectives. In contrast, the six-cluster solution fragmented the data into smaller segments without improving clustering quality, while  $K = 2$  produced only a coarse separation that offered limited business insight.

**Figure 7. Cluster Sizes and Profiles**



## 7.3 Cluster Profiles and Business Interpretation

### Cluster Sizes and Profiles

*Figure 7. (a) Cluster sizes with percentages. (b) Profile heatmap showing raw means (annotations) with column-normalized coloring.*

**Cluster 1.** Cluster 1 (n = 17169, 15.7%): Lowest mean review score (1.97) combined with moderate delivery duration (10.25 days), indicating that the available delivery variables do not fully explain dissatisfaction in this group.

**Cluster 2.** Cluster 2 (n = 10358, 9.5%): Below-average review score (2.70) with the longest average delivery duration (30.03 days) and positive delay (3.21 days). This cluster is labeled descriptively as the “delayed-delivery” group.

**Cluster 3.** Cluster 3 (n = 7388, 6.8%): Above-average review score (4.09) with notably higher item prices (R\$476.74 mean). This cluster is labeled the “premium-product” group.

**Cluster 4.** Cluster 4 (n = 74,451, 68.1%): Highest review score (4.76), shortest delivery duration (9.49 days), and earliest delivery relative to estimate (−13.81 days). The largest and highest-performing cluster.

**Important caveat.** Because review score was used as a clustering input, clusters are expected to differ on review scores by construction. This should not be cited as independent validation of the clustering solution. The more informative evidence is the separation in delivery duration, delivery delay, and item price, which were not directly optimized by the algorithm but show meaningful differentiation across clusters. Cluster labels (e.g., “delayed-delivery,” “premium-product”) are descriptive summaries, not confirmed customer types.

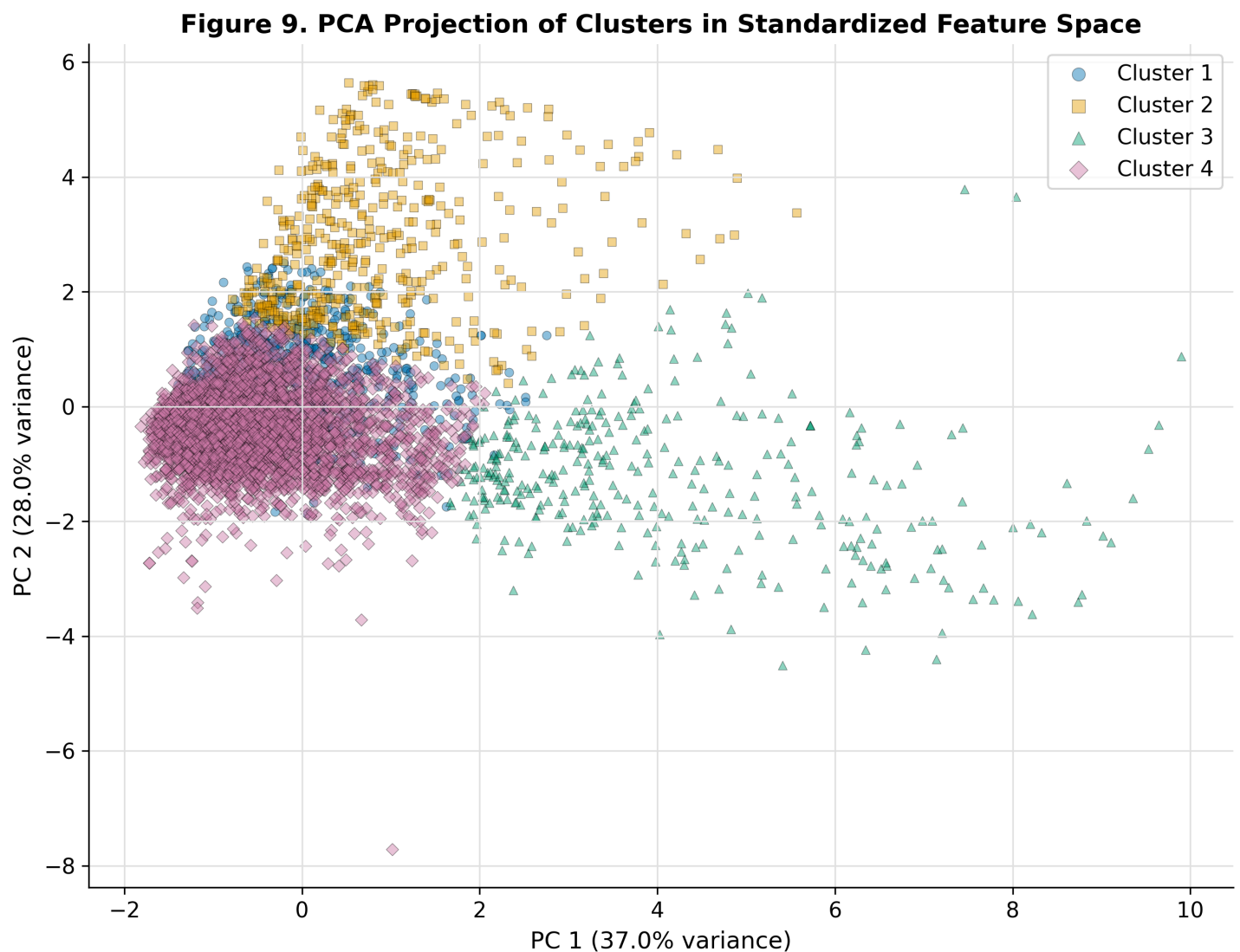
**Figure 8. Key Variables by Cluster**



## Key Variables by Cluster

**Figure 8. Boxplots of delivery duration, delivery delay, and review score by cluster. ANOVA result for review score:  $F = 70,765.5, p < 0.001$ .**

Cluster 2 displays a distinctly elevated distribution of both delivery duration and delivery delay. The ANOVA result ( $F = 70,765.5, p < 0.001$ ) confirms that review-score means differ across clusters, but this is a descriptive rather than validating finding because review score was a clustering input. The delivery variable separation provides stronger evidence of meaningful segmentation.



## PCA Cluster Visualization

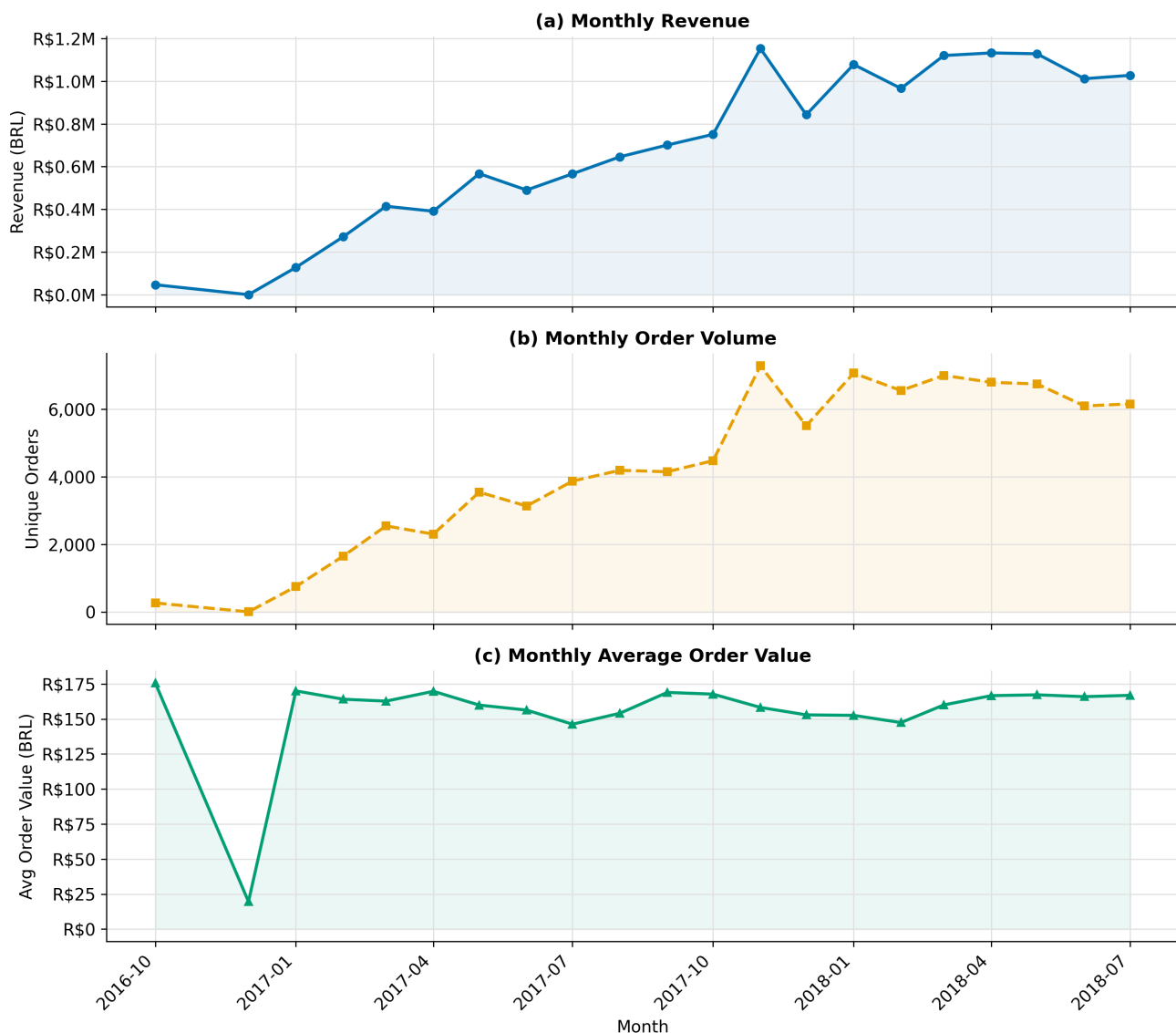
**Figure 9. PCA projection ( $PC1 = 37.0\%$ ,  $PC2 = 28.0\%$ ). Different markers distinguish clusters for grayscale accessibility.**

Cluster 2 (delayed deliveries) occupies a distinct region in the projection. Clusters 3 and 4 overlap substantially, consistent with differentiation primarily by item price rather than delivery variables. The moderate total variance explained (65.0%) is consistent with the modest silhouette scores reported in Figure 6.

### 7.4 Historical Sales Trends

The forecasting dataset uses delivered orders with payment values aggregated monthly. The first and last boundary months are excluded, producing 21 monthly observations (October 2016 through July 2018).

**Figure 10. Monthly Revenue, Order Volume, and Average Order Value**

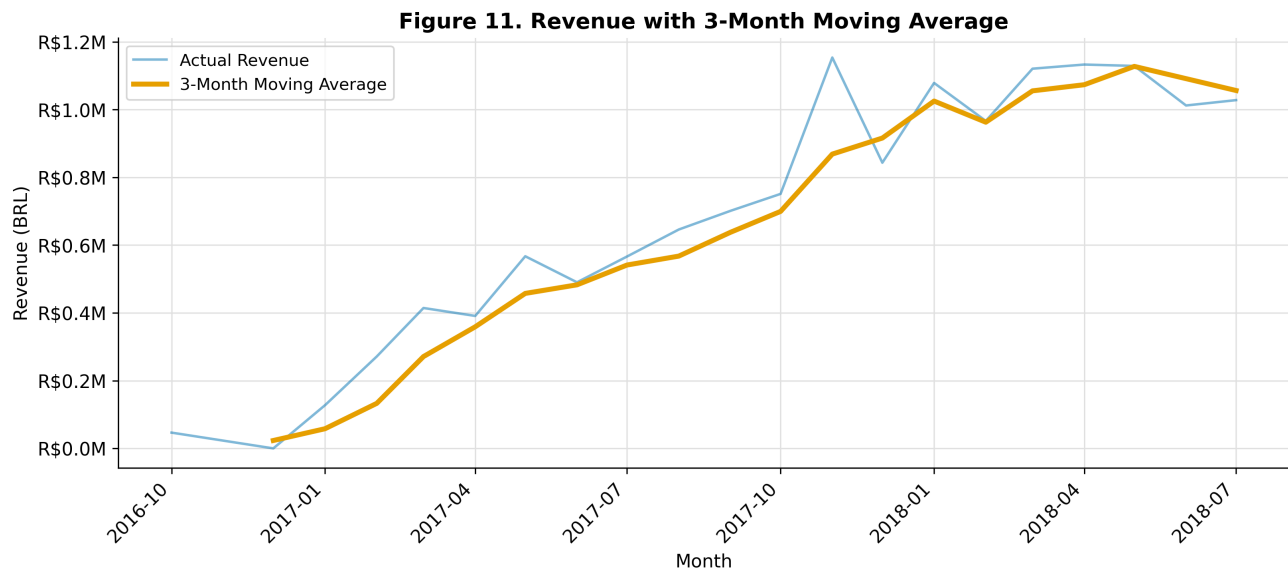


## Monthly Revenue, Order Volume, and Average Order Value

*Figure 10. Three aligned panels sharing the same time axis: (a) monthly revenue, (b) monthly unique orders, (c) monthly average order value.*

Panels (a) and (b) show that both revenue and order volume exhibit a clear upward trend across the observation period. Panel (c) adds the monthly average order value, calculated as total monthly revenue divided by the number of unique orders. Average Order Value declined slightly from approximately R\$176 to R\$167 (−5.0%), indicating that revenue growth was driven primarily by increasing order volume. This confirms that the upward revenue trend was driven primarily by increasing order volume rather than higher per-order spending.

### Revenue with Moving Average

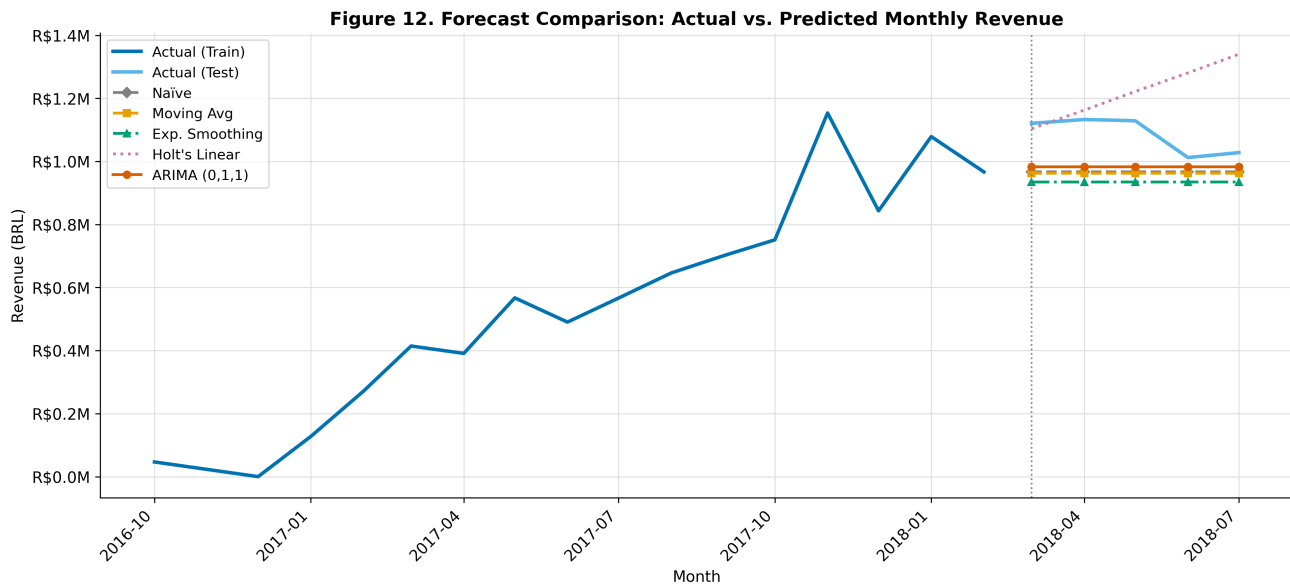


*Figure 11. Monthly revenue (light blue) with 3-month moving average (orange).*

The moving average rises consistently, confirming that the upward trend is persistent rather than driven by isolated high-revenue months.

## 7.5 Forecasting Model Evaluation

## Forecast Model Comparison



*Figure 12. Five models compared on a 5-month holdout test period. Train = dark blue, test = sky blue. Models distinguished by color, line style, and markers.*

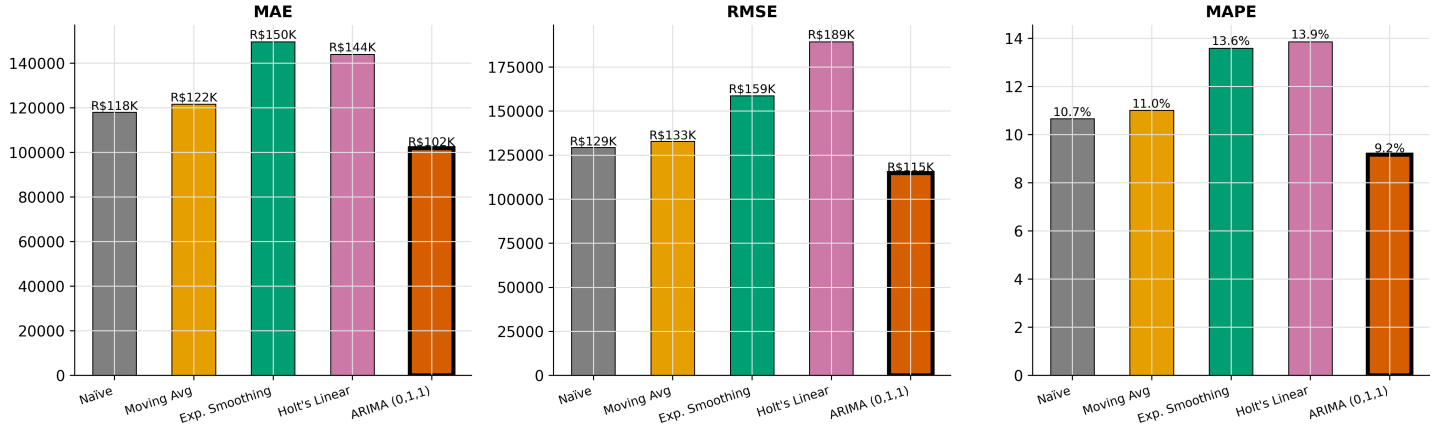
ARIMA (0,1,1) tracks the test period most closely and achieves the lowest forecast errors across MAE, RMSE, and MAPE. The naïve, moving average, and exponential smoothing methods produce flat forecasts that systematically underestimate revenue in a growing market. Forecast performance was evaluated over a 5-month historical test period.

### Forecast Accuracy

| Model          | MAE     | RMSE    | MAPE  |
|----------------|---------|---------|-------|
| Naïve          | 117,978 | 129,353 | 11.0% |
| Moving Avg     | 121,716 | 132,771 | 11.0% |
| Exp. Smoothing | 149,622 | 158,745 | 14.0% |
| Holt's Linear  | 144,037 | 189,337 | 14.0% |
| ARIMA (0,1,1)  | 101,903 | 114,881 | 9.2%  |

**Table 7. Forecast accuracy on the 5-month test period.**

**Figure 13. Forecast Accuracy by Model (Lower = Better; Bold Border = Best)**



**Figure 13. MAE, RMSE, and MAPE by model. Bold border = best model. Colors match Figure 12.**

ARIMA (0,1,1) achieves MAPE = 9.2% and reduces RMSE by approximately 11% relative to the naïve baseline.

*Limitation: The test period contains only 5 months. All models are univariate. The dataset ends in 2018; results should not be presented as current Olist forecasts.*

## 7.6 Overall Visualization Findings

**RQ1.** Delivery duration and reliability show modest but consistent negative associations with review scores ( $r = -0.30$  and  $-0.23$ , respectively). The K-Means solution at  $K = 4$  produces interpretable clusters: a delayed-delivery group (Cluster 2), a premium-product group (Cluster 3), a group where low satisfaction is not explained by delivery variables (Cluster 1), and a high-performing majority (Cluster 4). Because review score was a clustering input, the review-score differences across clusters are structural; the delivery and price separation provides the more meaningful evidence of segmentation quality.

**RQ2.** Monthly revenue exhibited an overall upward trend during the study period, driven primarily by increasing order volume, while the average order value declined slightly from approximately **RS\$176** to **RS\$167** ( $-5.0\%$ ). Among the five forecasting approaches evaluated,

**ARIMA (0,1,1)** produced the strongest predictive performance, achieving a **MAE of R\$101,903**, **RMSE of R\$114,881**, and **MAPE of 9.2%**. Compared with the naïve forecasting baseline, ARIMA reduced RMSE by approximately **11%**, making it the best-performing model for forecasting monthly sales revenue within the available historical data.

**Limitations.** The analyses identify statistical relationships and forecasting performance but do not establish causal relationships between variables. The observed correlations between delivery-related variables and customer review scores are modest and should be interpreted accordingly. In addition, the forecasting analysis is based on a relatively short time series of **21 monthly observations**, with model performance evaluated using a **5-month holdout test period**. Consequently, the forecasting results should be viewed as decision-support guidance rather than precise long-term predictions. All findings are based on historical Olist data collected between **2016 and 2018** and may not fully represent current e-commerce market conditions.

## **8. Ethical Recommendations**

### **8.1 Ethical Context**

An analysis of customer stratification, delivery performance, customer purchasing behavior, and sales forecasting within the Olist Brazilian E-Commerce Dataset raises several ethical considerations related to the use of business analytics in e-commerce. The project uses customer, order, payment, product, seller, review, and delivery data to identify satisfaction patterns and forecast future sales trends. The goal of this project is to improve customer experience, operational efficiency, inventory planning, and business decision-making within the e-commerce industry. However, these analytical findings should be used carefully because data-driven recommendations can influence how the business treats customers, sellers, product categories, and geographic regions. The variables considered across the project include review score, delivery time, freight value, payment type, order status, product category, and customer location; the completed clustering model discussed in Sections 6 and 7 narrows this list to six standardized numerical variables.

### **8.2 Customer Privacy and Responsible Data Use**

One important ethical issue involves customer privacy and responsible data use. Although the Olist dataset is public historical data, it still represents real customer transactions, seller activity, purchasing behavior, review ratings, and geographic patterns. Even though customers are not directly named, combining location, order, product, payment, and review data can still create privacy concerns if the resulting insights are used to target or profile specific customer groups too aggressively. The business should avoid using this type of analysis to manipulate customers, pressure them into unnecessary purchases, or exclude certain customer segments from promotions or services. Customer data should be analyzed in an aggregated way, and any reports or dashboards should focus on trends rather than attempting to identify individual customers.

### **8.3 Review-Score Bias and Missing Feedback**

A second ethical concern could be the possibility of bias in customer satisfaction ratings. The project uses “review\_score” as the primary measure of customer satisfaction, and the EDA shows that review scores are heavily concentrated around four- and five-star ratings. This can suggest that most customers reported positive experiences, but it can also mean that the model

may not fully capture the experiences of dissatisfied customers, especially if many of them did not leave written comments. The EDA also notes that the review comment titles and review messages had high missing values, so the analysis excluded those text fields and relied mainly on numerical scores.

#### **8.4 Seller and Regional Fairness in Delivery Decisions**

Another ethical issue involves how delivery-performance findings could affect sellers and logistics decisions. The EDA found negative relationships between review scores and delivery-related measures, including delivery days and delivery delay. Longer delivery times and late deliveries were associated with lower review scores, suggesting that delivery performance is a major factor in customer satisfaction. While this insight can be useful, the business should be careful not to unfairly punish sellers who operate in regions with weaker infrastructure, longer shipping distances, or fewer carrier options. If the company rewards only the sellers in areas with faster delivery networks, then smaller sellers or sellers in remote regions would be at a disadvantage. The ethical use of this research should separate seller performance from structural logistical challenges and should focus on improving delivery systems rather than penalizing sellers.

#### **8.5 Responsible Cluster Interpretation**

The clustering portion of the project can also raise ethical concerns. The project's clustering work was originally framed around review score, delivery date, delivery delay, price, freight value, payment value, product category, seller state, order status, and payment type, although the completed K-Means model used six standardized numerical variables, as described in Section 6.1. While clustering can help identify meaningful customer-experience patterns, it can also lead to oversimplified labels. For example, a cluster with low review scores and long delivery times should not be automatically interpreted as "bad customers," "bad sellers," or "bad regions." Instead, clusters should be interpreted as operational patterns that help the business understand where service can be improved. Cluster labels should be neutral and business-focused, such as "high-delay" or "high-dissatisfaction" orders, and "low-delay" or "high-satisfaction" orders.

## **8.6 Forecasting, Inventory, and Product Diversity**

Sales forecasting creates another ethical challenge because forecast results can influence inventory planning, marketing, staffing, and seller support. If the business relies too heavily on forecasting models, it may overstock items that are not needed, understock other products, or prioritize only the categories predicted to generate higher revenue. This could reduce product diversity and limit opportunities for smaller sellers. Forecasting should support business planning, but it should not be the only factor used when making these decisions.

## **8.7 Regional and Category Fairness**

There are also ethical risks related to regional and category-based recommendations. The project studies purchasing behavior across product categories and geographic regions, including regional sales trends, high-performing categories, and seasonal purchasing. These insights can help businesses improve marketing campaigns and inventory planning, but they can also lead to unequal treatment if certain regions or product categories are viewed as less profitable. For example, customers in lower-sales regions should not receive worse service, fewer promotions, or slower delivery support simply because their area is less profitable. Ethical business recommendations should focus on improving access, service quality, and operational performance across regions, not just maximizing revenue from the highest-performing locations.

## **8.8 Risks of Ignoring the Findings**

The potential impact of not using this research should also be considered. If Olist or similar e-commerce businesses ignore customer satisfaction and delivery performance data, they may continue to experience preventable service problems, delivery delays, inefficient inventory planning, and customer dissatisfaction. The project's expected outputs include dashboards and customer segmentation designed to improve customer experience, operational efficiency, and sales performance. Not using these insights could result in missed opportunities to improve delivery reliability, reduce negative reviews, support sellers, and better prepare for periods of high customer demand.

## **8.9 Transparency, Human Oversight, and Responsible Use**

Overall, these findings should be used as decision-support tools rather than automatic decision-making systems. The business should avoid relying only on review scores, clusters, or forecasts when making strategic decisions. Instead, they should combine model results with

human judgment, seller feedback, customer service information, and awareness of regional logistics limitations. The project's recommendations should emphasize transparency, fairness, privacy protection, and responsible interpretation. Used ethically, this research can help the e-commerce business improve customer experience, strengthen delivery operations, support sellers, and plan for future demand without unfairly disadvantaging customers, regions, or product categories.

## 9. Challenges and Limitations

**Preliminary versus completed dataset construction.** The EDA in Section 5 was based on an earlier 119,140-observation merge, while the completed analysis in Section 7 used a revised 113,425-record, 99,441-order merge (see Section 5.8). This difference in dataset construction should be kept in mind when comparing EDA and Section 7 findings.

**Relational-table duplication risks.** Orders can contain multiple items and multiple payment transactions. Repeated review scores in multi-item orders inflate absolute order-item counts even when the distributional shape remains similar. The final analysis reduced these risks by aggregating payments at the order level and deduplicating reviews to one record per order before joining to order items.

**Missing review and delivery information.** Review-comment titles were 88.34% missing and review-comment messages were 58.70% missing, so the analysis relied primarily on numerical review scores. Delivery-related dates also contained smaller amounts of missing information for canceled, unavailable, or undelivered orders.

**Skewed variables and outliers.** Price, freight value, payment value, and delivery duration were strongly right-skewed. The EDA retained valid extreme transactions, while the completed clustering model capped its six input variables at the 99th percentile. This difference should remain clear when interpreting the EDA and model results.

**Modest cluster separation.** The selected four-cluster solution had a silhouette score of 0.310, while  $K = 6$  reached 0.243. The PCA projection explained 65% of total variance, and several clusters overlapped. Cluster labels are descriptive summaries rather than confirmed customer types.

**Structural review-score differences.** Review score was used as a clustering input; therefore, differences in review scores across clusters are expected by construction and cannot independently validate the cluster solution. Delivery and price differences are more informative for interpretation.

**Correlation does not establish causation.** The final correlations between review score and delivery duration (-0.30) and delivery delay (-0.23) were modest. Product type, geography, seller quality, or other omitted variables may contribute to the observed relationships.

**Limited forecasting history and evaluation window.** The forecasting series contains 21 monthly observations and only five test months. All reported forecasting models are univariate,

and the dataset ends in 2018. The reported errors should not be treated as current forecasts or as evidence of performance in other periods.

**Generalizability.** Findings are limited to the historical Olist dataset and the variables included in the submitted analysis. Results may not represent current platform conditions or other e-commerce markets.

## **10. Recommendations and Next Steps**

### **10.1 Immediate Operational Recommendations**

- Track delivery duration and delivery delay as routine operational indicators, with particular attention to the delayed-delivery cluster, which averaged 30.03 delivery days and a +3.21-day delay.
- Use cluster profiles as diagnostic patterns rather than automatic classifications of customers, sellers, or regions.

### **10.2 Customer-Satisfaction Recommendations**

- Prioritize the reduction of late and extended deliveries because the completed analysis shows modest but consistent negative relationships between delivery performance and review scores.
- Investigate the low-satisfaction cluster whose dissatisfaction was not explained by the available delivery variables. Additional customer-service, product-quality, and qualitative feedback may be needed before taking action.

### **10.3 Delivery and Logistics Recommendations**

- Focus on improving delivery systems and estimated-delivery accuracy rather than automatically penalizing sellers. Regional infrastructure, shipping distance, and carrier availability should be considered when evaluating performance.
- Use neutral operational labels such as delayed-delivery or high-satisfaction orders and avoid labels that imply bad customers, bad sellers, or bad regions.

### **10.4 Forecasting and Inventory-Planning Recommendations**

- Use ARIMA (0,1,1) as the best-performing model within the submitted five-month holdout comparison, while treating it as decision support rather than a guaranteed projection.
- Update and re-evaluate forecasting models with more recent data before using them for inventory, staffing, marketing, or seller-support decisions. Revenue growth in the submitted period was driven mainly by order volume rather than rising average order value.
- Avoid relying on forecasts alone when allocating inventory. Consider product diversity, smaller-seller opportunities, and the risk of overstocking or understocking.

## **10.5 Responsible-Use Safeguards**

- Analyze customer data in aggregate and avoid aggressive profiling, manipulation, exclusion from promotions, or attempts to identify individuals.
- Combine model outputs with human judgment, seller feedback, customer-service information, and awareness of regional logistics limitations.
- Maintain transparency about review-score bias, missing text feedback, cluster uncertainty, forecast error, and the historical scope of the dataset.

## **10.6 Future Analytical Work**

- Evaluate the categorical variables originally proposed in the methodology, because the completed K-Means model used only six numerical features.
- Assess additional forecasting approaches, including seasonal ARIMA, SARIMA, Prophet, or machine-learning-based forecasting models, using a longer historical time series and more recent data.

## 11. Conclusion

For RQ1, the completed analysis indicates that K-Means can identify four interpretable groups based on customer satisfaction and operational characteristics, but the separation is modest and should not be overstated. The most meaningful operational pattern is the distinction between the delayed-delivery cluster and the high-performing majority cluster. Review-score differences are partly structural because the review score was an input, and one low-satisfaction cluster shows that delivery variables alone do not explain every negative customer experience.

For RQ2, historical monthly revenue can be forecast with useful accuracy within the submitted holdout test. ARIMA (0,1,1) achieved the best forecasting performance, with RMSE of R\$114,881 and MAPE of 9.2%. The model reduced forecasting error by approximately 11% relative to the naïve baseline and demonstrated the ability to capture temporal patterns in historical sales revenue. These results apply only to the 2016-2018 historical period and a five-month test window. The analysis, therefore, supports trend-aware operational planning, but forecasts should be updated with current data and used alongside human judgment, fairness safeguards, and uncertainty disclosure.

## References

Olist Brazilian E-Commerce Public Dataset. Kaggle. Retrieved from: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Wong, B. (2011). Points of view: Color blindness. *Nature Methods*, 8(6), 441.

## Appendix A: Source Code

The following code is reproduced from the submitted Jupyter Notebook. It was not re-executed or modified for this report.

### Library Imports and Configuration

```
"""
Olist Data Visualization — FINAL version
All 16 corrections applied. Wong colorblind palette.
"""

import os, warnings, json, numpy as np, pandas as pd
import matplotlib.pyplot as plt, matplotlib.ticker as mt
import matplotlib.patches as mp
from statsmodels.tsa.arima.model import ARIMA
from matplotlib.lines import Line2D
import seaborn as sns
from scipy import stats
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA
warnings.filterwarnings("ignore")

DATA="./data"; OUT="./final_visualizations"; os.makedirs(OUT, exist_ok=True)

# Wong palette
W_BL="#0072B2"; W_OR="#E69F00"; W_GR="#009E73"; W_SB="#56B4E9"
W_VR="#D55E00"; W_PU="#CC79A7"; W_YL="#F0E442"; W_BK="#000000"
CL=[W_BL, W_OR, W_GR, W_PU]
STAR_C=[W_VR, W_OR, W_YL, W_SB, W_BK]

plt.rcParams.update({"figure.facecolor": "white", "axes.facecolor": "white",
    "axes.edgecolor": "black", "axes.labelcolor": "black", "text.color": "black",
    "xtick.color": "black", "ytick.color": "black", "axes.spines.top": False,
    "axes.spines.right": False, "axes.grid": True, "grid.color": "#E0E0E0",
    "font.family": "DejaVu Sans", "font.size": 11, "axes.titlesize": 13,
    "axes.titleweight": "bold", "figure.dpi": 150, "savefig.dpi": 300,
    "savefig.facecolor": "white", "savefig.bbox": "tight"})

def sv(n):
    p = os.path.join(OUT, n)
    plt.savefig(p, dpi=300, bbox_inches="tight", facecolor="white")
```

```

plt.show()
print(f"✓ {n}")
plt.close("all")

# ——— DATA LOADING & JOIN AUDIT (Correction #1) ———

def ld(f):return pd.read_csv(os.path.join(DATA,f))
print("=== JOIN AUDIT ===")
orders=ld("olist_orders_dataset.csv")
reviews=ld("olist_order_reviews_dataset.csv")
payments=ld("olist_order_payments_dataset.csv")
items=ld("olist_order_items_dataset.csv")
products=ld("olist_products_dataset.csv")
sellers=ld("olist_sellers_dataset.csv")
try:
    ct=ld("product_category_name_translation.csv")
    products=products.merge(ct,on="product_category_name",how="left")
    products["product_category_name"]=products["product_category_name_english"].fillna(products["product_category_name"])
except:pass

audit=[]

def log(name,before,key,jtype,rel,after):
    unm=before-after if after<before else 0
    audit.append({"Table":name,"Rows Before":before,"Key":key,"Join":jtype,
        "Relationship":rel,"Rows After":after,"Unmatched":unm})
    print(f" {name}: {before:,} → {after:,} ({jtype} on {key})")

# Date conversion + feature engineering
for c in ["order_purchase_timestamp","order_delivered_customer_date","order_estimated_delivery_date"]:
    orders[c]=pd.to_datetime(orders[c],errors="coerce")
orders["delivery_days"]=(orders["order_delivered_customer_date"]-orders["order_purchase_timestamp"]).dt.days
orders["delivery_delay"]=(orders["order_delivered_customer_date"]-orders["order_estimated_delivery_date"]).dt.days

# Step 1: Aggregate payments
n0=len(payments)
pay_agg=payments.groupby("order_id")["payment_value"].sum().reset_index()
n1=len(pay_agg)
dup_pay=payments.order_id.duplicated().sum()
log("payments→agg",n0,"order_id","groupby","many→1",n1)
print(f" Dup order_ids in payments: {dup_pay:,}")

# Step 2: Deduplicate reviews
n0r=len(reviews)

```

```

dup_rev=reviews.order_id.duplicated().sum()
reviews_d=reviews.drop_duplicates(subset="order_id",keep="first")
n1r=len(reviews_d)
log("reviews→dedup",n0r,"order_id","dedup","many→1",n1r)
print(f"  Dup order_ids in reviews: {dup_rev:,}")

# Step 3: Build analytical dataset
n_ord=len(orders)
df=orders.merge(reviews_d[["order_id","review_score"]],on="order_id",how="left")
log("orders+reviews",n_ord,"order_id","left 1:1","1:1",len(df))

df=df.merge(pay_agg,on="order_id",how="left")
log("+payments_agg",len(df),"order_id","left 1:1","1:1",len(df))

n_pre_items=len(df)
df=df.merge(items[["order_id","product_id","seller_id","price","freight_value"]],on="order_id",how="left")
log("+items",n_pre_items,"order_id","left 1:many","1:many",len(df))

df=df.merge(products[["product_id","product_category_name"]],on="product_id",how="left")
log("+products",len(df),"product_id","left","many:1",len(df))

df=df.merge(sellers[["seller_id","seller_state"]],on="seller_id",how="left")
log("+sellers",len(df),"seller_id","left","many:1",len(df))

TOTAL_ROWS=len(df)
UNIQUE_ORDERS=df.order_id.nunique()
print(f"\n  FINAL: {TOTAL_ROWS:,} rows, {UNIQUE_ORDERS:,} unique orders")
audit_df=pd.DataFrame(audit)
print(audit_df.to_string(index=False))

# ===== EXACT STATISTICS (Corrections #2, #3) =====
# Correlations from individual records
corr_all=df[["review_score","price","freight_value","payment_value","delivery_days","delivery_delay"]].dropna().corr()
r_score_days=corr_all.loc["review_score","delivery_days"]
r_score_delay=corr_all.loc["review_score","delivery_delay"]
r_days_delay=corr_all.loc["delivery_days","delivery_delay"]
r_price_pay=corr_all.loc["price","payment_value"]
print(f"\n=== CORRELATIONS (individual records, 2dp) ===")
print(f"  review_score & delivery_days: {r_score_days:.2f}")
print(f"  review_score & delivery_delay: {r_score_delay:.2f}")
print(f"  delivery_days & delivery_delay:{r_days_delay:.2f}")
print(f"  price & payment_value:      {r_price_pay:.2f}")

```

```

# Delivery timing (Correction #3)
dl_all=df["delivery_delay"].dropna()
pct_early=(dl_all<0).mean()*100
pct_ontime=(dl_all==0).mean()*100
pct_late=(dl_all>0).mean()*100
print(f"\n=== DELIVERY TIMING ===")
print(f" Early (delay<0): {pct_early:.1f}%")
print(f" On-time (delay=0): {pct_ontime:.1f}%")
print(f" Late (delay>0): {pct_late:.1f}%")
print(f" Total: {pct_early+pct_ontime+pct_late:.1f}%")

# Review stats
rm=df.review_score.mean();rmd=df.review_score.median()
print(f"\n Review: mean={rm:.2f}, median={rmd:.0f}")

# ——— RQ2 time-series + AOV (Correction #4) ———
ts=orders.merge(pay_agg,on="order_id",how="left")
ts=ts[ts.order_status=="delivered"].copy()
ts["month"]=ts.order_purchase_timestamp.dt.to_period("M")
mrev=ts.groupby("month")["payment_value"].sum().sort_index().iloc[1:-1]
mord=ts.groupby("month")["order_id"].nunique().sort_index().iloc[1:-1]
maov=mrev/mord # Average Order Value
rv=mrev.to_timestamp();od=mord.to_timestamp();av=maov.to_timestamp()
print(f"\n=== TIME SERIES ===")
print(f" Months: {len(mrev)}, Range: {mrev.index[0]}—{mrev.index[-1]}")
print(f" AOV range: R${av.min():.0f} – R${av.max():.0f}")
print(f" AOV trend: {'increasing' if av.iloc[-1]>av.iloc[0] else 'stable/decreasing'}")

# ——— FIGURE 1: Review Distribution ———
print("\n=== FIGURES ===")
print("Fig 1")
sc=df.review_score.value_counts().sort_index();pct=sc/sc.sum()*100
fig,ax=plt.subplots(figsize=(7.5,5))
bars=ax.bar(sc.index,sc.values,color=STAR_C,edgecolor="black",lw=0.6,width=0.6)
for b,p in zip(bars,pct):
    ax.text(b.get_x()+b.get_width()/2,b.get_height()+400,f"{p:.1f}%",ha="center",fontsize=10)
ax.set_xlabel("Review Score (Stars)",fontsize=11);ax.set_ylabel("Count",fontsize=11)
ax.set_xticks([1,2,3,4,5]);ax.set_title("Figure 1. Distribution of Customer Review Scores")
ax.yaxis.set_major_formatter(met.FuncFormatter(lambda x,_:f"{int(x):,}"))
ax.set_ylim(0,sc.max()*1.18);plt.tight_layout();sv("figure_01.png")

```

# ——— FIGURE 2: Delivery Distributions ———

```
print("Fig 2")
dd=df.delivery_days.dropna();dl=df.delivery_delay.dropna()
fig,axes=plt.subplots(1,2,figsize=(14,5))
ax=axes[0]
ax.hist(dd[(dd>=0)&(dd<=60)],bins=40,color=W_BL,edgecolor="white",lw=0.3,alpha=0.85)
ax.axvline(dd.median(),color=W_BK,ls="--",lw=2,label=f"Median: {dd.median():.0f} days")
ax.axvline(dd.mean(),color=W_VR,ls=".",lw=2,label=f"Mean: {dd.mean():.1f} days")
ax.set_xlabel("Delivery Duration (Days)",fontsize=11);ax.set_ylabel("Frequency",fontsize=11)
ax.set_title("(a) Delivery Duration",fontsize=12);ax.legend(fontsize=10)
ax.yaxis.set_major_formatter(mt.FuncFormatter(lambda x, _:f"{int(x):,}"))
ax=axes[1]
dlc=dl[(dl>=dl.quantile(0.01))&(dl<=dl.quantile(0.99))]
ax.hist(dlc,bins=50,color=W_GR,edgecolor="white",lw=0.3,alpha=0.85)
ax.axvline(0,color=W_BK,lw=2.5,label="On-time (0 days)")
ax.axvline(dl.median(),color=W_VR,ls="--",lw=2,label=f"Median: {dl.median():.1f} days")
ax.set_xlabel("Delivery Delay (Negative = Early)",fontsize=11);ax.set_ylabel("Frequency",fontsize=11)
ax.set_title("(b) Delivery Delay",fontsize=12);ax.legend(fontsize=10)
ax.yaxis.set_major_formatter(mt.FuncFormatter(lambda x, _:f"{int(x):,}"))
fig.suptitle("Figure 2. Delivery Performance Distributions",fontweight="bold",fontsize=13,y=1.01)
plt.tight_layout();sv("figure_02.png")
```

# ——— FIGURE 3: Review vs Delivery (consistent r values) ———

```
print("Fig 3")
fig,axes=plt.subplots(1,2,figsize=(14,5.5))
s1=df[["review_score","delivery_days"]].dropna()
s1=s1[(s1.delivery_days>=0)&(s1.delivery_days<=60)].copy()
r1_ind,_=stats.pearsonr(s1.delivery_days,s1.review_score)
s1["bin"]=(s1.delivery_days//5)*5
g1=s1.groupby("bin").review_score.agg(["mean","count","sem"]).reset_index().query("count>=30")
ax=axes[0]
ax.errorbar(g1.bin,g1["mean"],yerr=1.96*g1["sem"],fmt="o-",color=W_BL,lw=2,ms=7,capsize=5,elinewidth=1.2,label="Mean ± 95% CI")
m,b=np.polyfit(s1.delivery_days,s1.review_score,1);xl=np.linspace(0,60,100)
ax.plot(xl,m*xl+b,"--",color=W_VR,lw=1.8,label=f"Trend (r = {r_score_days:.2f})")
ax.set_xlabel("Delivery Duration (5-Day Bins)",fontsize=11);ax.set_ylabel("Mean Review Score",fontsize=11)
ax.set_title("(a) By Delivery Duration",fontsize=12);ax.set_ylim(1,5.5);ax.set_yticks([1,2,3,4,5])
ax.legend(fontsize=10)

s2=df[["review_score","delivery_delay"]].dropna()
s2=s2[(s2.delivery_delay>=-30)&(s2.delivery_delay<=30)].copy()
r2_ind,_=stats.pearsonr(s2.delivery_delay,s2.review_score)
s2["bin"]=(s2.delivery_delay//5)*5
```

```

g2=s2.groupby("bin").review_score.agg(["mean","count","sem"]).reset_index().query("count>=30")
ax=axes[1]
ax.errorbar(g2.bin,g2["mean"],yerr=1.96*g2["sem"],fmt="s-",color=W_GR,lw=2,ms=7,capsize=5,elinewidth=1.2,label="Mean ± 95% CI")
m2,b2=np.polyfit(s2.delivery_delay,s2.review_score,1);x2=np.linspace(-30,30,100)
ax.plot(x2,m2*x2+b2,"--",color=W_VR,lw=1.8,label=f"Trend (r = {r_score_delay:.2f})")
ax.axvline(0,color="gray",ls=":",lw=1.5)
ax.set_xlabel("Delivery Delay (5-Day Bins)",fontsize=11);ax.set_ylabel("Mean Review Score",fontsize=11)
ax.set_title("(b) By Delivery Delay",fontsize=12);ax.set_ylim(1,5.5);ax.set_yticks([1,2,3,4,5])
ax.legend(fontsize=10)
fig.suptitle("Figure 3. Mean Review Score by Delivery Performance",fontweight="bold",fontsize=13,y=1.01)
plt.tight_layout();sv("figure_03.png")
# NOTE: r values in legend use corr_all (full dataset), consistent with Fig 4

```

# ——— FIGURE 4: Correlation Heatmap ———

```

print("Fig 4")
cv=["review_score","price","freight_value","payment_value","delivery_days","delivery_delay"]
nl=["Review\nScore","Item\nPrice","Freight\nValue","Payment\nValue","Delivery\nDuration","Delivery\nDelay"]
fig,ax=plt.subplots(figsize=(7.5,6.5))
im=ax.imshow(corr_all.values,cmap="RdBu_r",vmin=-1,vmax=1,aspect="auto")
for i in range(len(cv)):
    for j in range(len(cv)):
        v=corr_all.iloc[i,j];tc="white" if abs(v)>0.55 else "black"
        ax.text(j,i,f"{v:.2f}",ha="center",va="center",fontsize=10,color=tc,fontweight="bold")
ax.set_xticks(range(len(cv)));ax.set_xticklabels(nl,fontsize=9)
ax.set_yticks(range(len(cv)));ax.set_yticklabels(nl,fontsize=9)
plt.colorbar(im,ax=ax,fraction=0.046,pad=0.04).set_label("Pearson r",fontsize=10)
ax.set_title("Figure 4. Correlation Matrix of Key Variables")
plt.tight_layout();sv("figure_04.png")

```

# ——— FIGURE 5: Categories Top+Bottom ———

```

print("Fig 5")
cs=df.groupby("product_category_name").review_score.agg(mean="mean",count="count").query("count>=200").sort_values("mean")
top10=cs.tail(10);bot10=cs.head(10);show=pd.concat([bot10,top10])
oa=df.review_score.mean()
fig,ax=plt.subplots(figsize=(10.5,7.5))
yp=range(len(show))
colors=[W_BL if v>=oa else W_VR for v in show["mean"]]
bars=ax.barh(list(yp),show["mean"].values,color=colors,edgecolor="black",lw=0.5,height=0.65)
ax.set_yticks(list(yp));ax.set_yticklabels([c.replace("_"," ").title() for c in show.index],fontsize=9)
ax.axvline(oa,color=W_BK,ls="--",lw=1.5)
for bb,v,n in zip(bars,show["mean"],show["count"]):
    ax.text(v+0.01,bb.get_y()+bb.get_height()/2,f"{v:.2f} (n={n:})",va="center",ha="left",fontsize=8)

```

```

ax.set_xlim(2,5.5);ax.set_xlabel("Mean Review Score",fontsize=11)
ax.set_title("Figure 5. Mean Review Score: Top & Bottom 10 Categories (Min. 200 Records)")
ax.legend(handles=[mp.Patch(fc=W_BL,ec="black",label="≥ Overall mean"),
    mp.Patch(fc=W_VR,ec="black",label="< Overall mean"),
    Line2D([0],[0],color="black",ls="--",lw=1.5,label=f"Overall mean: {oa:.2f}")),fontsize=9)
plt.tight_layout();sv("figure_05.png")

# ——— CLUSTERING ———

cf=["review_score","delivery_days","delivery_delay","price","freight_value","payment_value"]
cdf=df[cf].dropna().copy()
for c in cf:cdf[c]=cdf[c].clip(upper=cdf[c].quantile(0.99))
scaler=StandardScaler();X=scaler.fit_transform(cdf)
N_CLUST=len(cdf)
print(f"\nClustering: {N_CLUST:,} rows")

# ——— FIGURE 6: K-Selection ———

print("Fig 6")
wcss=[]
for k in range(1,11):
    km=KMeans(n_clusters=k,random_state=42,n_init=10);km.fit(X);wcss.append(km.inertia_)
np.random.seed(42);si=np.random.choice(len(X),min(8000,len(X)),replace=False);Xs=X[si]
sil={}
for k in range(2,11):
    km=KMeans(n_clusters=k,random_state=42,n_init=10);lb=km.fit_predict(Xs)
    sil[k]=silhouette_score(Xs,lb)
# Stability (Correction #5)
stab={}
for k in [2,3,4,5,6]:
    scores=[]
    for seed in [42,123,456,789,1010]:
        km=KMeans(n_clusters=k,random_state=seed,n_init=10);lb=km.fit_predict(Xs)
        scores.append(silhouette_score(Xs,lb))
    stab[k]={ "mean":np.mean(scores),"std":np.std(scores)}

bks=max(sil,key=sil.get)
sil4=sil[4];sil6=sil[6];sil_diff=sil6-sil4
print(f" Silhouette K=4: {sil4:.3f}, K=6: {sil6:.3f}, diff: {sil_diff:.3f}")

fig,axes=plt.subplots(1,2,figsize=(14,5.5))
axes[0].plot(range(1,11),wcss,"o-",color=W_BL,lw=2.5,ms=8)
axes[0].set_xlabel("Number of Clusters (K)",fontsize=11);axes[0].set_ylabel("WCSS",fontsize=11)
axes[0].set_title("(a) Elbow Method",fontsize=12);axes[0].set_xticks(range(1,11))

```

```

axes[0].yaxis.set_major_formatter(mt.FuncFormatter(lambda x, _:f'{x:,.0f}'))
ks=list(range(2,11));ss=[sil[k] for k in ks]
bars=axes[1].bar(ks,ss,color=[W_OR if k==bks else W_SB for k in ks],edgecolor="black",lw=0.6,width=0.6)
for bb,v in zip(bars,ss):
    axes[1].text(bb.get_x()+bb.get_width()/2,bb.get_height()+0.003,f'{v:.3f}',ha="center",fontsize=9)
axes[1].set_xlabel("Number of Clusters (K)",fontsize=11);axes[1].set_ylabel("Avg Silhouette Score",fontsize=11)
axes[1].set_title("(b) Silhouette Analysis",fontsize=12);axes[1].set_xticks(ks)
fig.suptitle("Figure 6. Cluster Number Selection Diagnostics",fontweight="bold",fontsize=13,y=1.01)
plt.tight_layout();sv("figure_06.png")

# Fit final K=4
OK=4
km_f=KMeans(n_clusters=OK,random_state=42,n_init=20,max_iter=500)
cdf=cdf.copy();cdf["cr"]=km_f.fit_predict(X)
om=(cdf.groupby("cr").review_score.mean().sort_values().reset_index()
    .assign(cluster=lambda d:range(1,OK+1)).set_index("cr")["cluster"].to_dict())
cdf["cluster"]=cdf.cr.map(om)
cmeans=cdf.groupby("cluster")[cf].mean()
csizes=cdf.cluster.value_counts().sort_index()
print(f" Sizes: {csizes.to_dict()}")
print(cmeans.round(2).to_string())

# ——— FIGURE 7: Cluster Sizes + Profile (bigger text) ———
print("Fig 7")
fig,axes=plt.subplots(1,2,figsize=(15,5.5),gridspec_kw={"width_ratios":[1,2.2]})
ax=axes[0]
bars=ax.bar([f"C{i}" for i in csizes.index],csizes.values,
    color=[CL[i-1] for i in csizes.index],edgecolor="black",lw=0.7)
for bb,v in zip(bars,csizes.values):
    ax.text(bb.get_x()+bb.get_width()/2,bb.get_height()+100,
        f'{v:,}\n({v/csizes.sum()*100:.1f}%)',ha="center",fontsize=9)
ax.set_ylabel("Records",fontsize=11);ax.set_title("(a) Cluster Sizes",fontsize=12)
ax.yaxis.set_major_formatter(mt.FuncFormatter(lambda x, _:f'{int(x):,}'));ax.set_ylim(0,csizes.max()*1.28)
ax=axes[1]
disp=cmeans.copy();disp.index=[f"C{i}" for i in disp.index]
disp.columns=["Review\nScore","Delivery\nDuration","Delivery\nDelay","Item\nPrice","Freight\nValue","Payment\nValue"]
norm=(disp-disp.min())/(disp.max()-disp.min()+1e-9)
annot=disp.round(1).astype(str)
sns.heatmap(norm,annot=annot,fmt="",cmap="RdYlBu_r",linewidths=1,linecolor="white",
    cbar_kws={"label":"Normalized"},annot_kws={"size":11,"color":"black"},ax=ax)
ax.set_title("(b) Cluster Profiles (Raw Means)",fontsize=12)
plt.yticks(rotation=0,fontsize=10);plt.xticks(fontsize=9)

```

```
fig.suptitle("Figure 7. Cluster Sizes and Profiles",fontweight="bold",fontsize=13,y=1.01)
plt.tight_layout();sv("figure_07.png")
```

# ——— FIGURE 8: Cluster Boxplots ———

```
print("Fig 8")
fig,axes=plt.subplots(1,3,figsize=(15,5.5))
for ax,var,title in zip(axes,["delivery_days","delivery_delay","review_score"],
    ["Delivery Duration (Days)","Delivery Delay (Days)","Review Score"]):
    dl=[cdf[cdf.cluster==i][var] for i in sorted(cdf.cluster.unique())]
    bp=ax.boxplot(dl,patch_artist=True,medianprops=dict(color="black",lw=2.5),
        flierprops=dict(marker="o",ms=2,alpha=0.15,mfc="gray",mec="gray"),
        whiskerprops=dict(lw=1.5),capprops=dict(lw=1.5))
    for pp,c in zip(bp["boxes"],CL[:OK]):pp.set_facecolor(c);pp.set_alpha(0.7)
    ax.set_xticklabels([f"C{i}" for i in sorted(cdf.cluster.unique())],fontsize=10)
    ax.set_title(title,fontsize=11)
    if var=="review_score":ax.set_yticks([1,2,3,4,5])
groups=[cdf[cdf.cluster==i].review_score.values for i in sorted(cdf.cluster.unique())]
fstat,pval=stats.f_oneway(*groups)
axes[2].text(0.97,0.97,f"ANOVA F={fstat:.1f}\np < 0.001",
    transform=axes[2].transAxes,ha="right",va="top",fontsize=10,
    bbox=dict(boxstyle="round",facecolor="lightyellow",edgecolor="gray"))
fig.suptitle("Figure 8. Key Variables by Cluster",fontweight="bold",fontsize=13,y=1.01)
plt.tight_layout();sv("figure_08.png")
print(f" ANOVA F={fstat:.1f}")
```

# ——— FIGURE 9: PCA (improved visibility) ———

```
print("Fig 9")
pca=PCA(n_components=2,random_state=42);Xpca=pca.fit_transform(X)
ev=pca.explained_variance_ratio_
np.random.seed(42);pidx=np.random.choice(len(Xpca),min(5000,len(Xpca)),replace=False)
markers=["o","s","^","D"]
fig,ax=plt.subplots(figsize=(9,7))
for ci in sorted(cdf.cluster.unique()):
    mask=cdf.cluster.values==ci;idx=np.intersect1d(np.where(mask)[0],pidx)
    ax.scatter(Xpca[idx,0],Xpca[idx,1],c=CL[ci-1],marker=markers[ci-1],
        alpha=0.45,s=20,label=f"Cluster {ci}",edgecolors="black",linewidth=0.3)
ax.set_xlabel(f"PC 1 ({ev[0]*100:.1f}% variance)",fontsize=12)
ax.set_ylabel(f"PC 2 ({ev[1]*100:.1f}% variance)",fontsize=12)
ax.set_title("Figure 9. PCA Projection of Clusters in Standardized Feature Space",fontsize=13)
ax.legend(fontsize=11,markerscale=1.5)
plt.tight_layout();sv("figure_09.png")
print(f" PC1={ev[0]*100:.1f}%, PC2={ev[1]*100:.1f}%")
```

# ——— FIGURE 10: 3-Panel (Revenue, Orders, AOV) — Correction #8 ———

```
print("Fig 10")
fig, axes = plt.subplots(3, 1, figsize=(11, 10), sharex=True)
axes[0].plot(rv.index, rv.values, "o-", color=W_BL, lw=2, ms=5)
axes[0].fill_between(rv.index, rv.values, alpha=0.08, color=W_BL)
axes[0].set_ylabel("Revenue (BRL)", fontsize=11)
axes[0].set_title("(a) Monthly Revenue", fontsize=12)
axes[0].yaxis.set_major_formatter(mtt.FuncFormatter(lambda x, _ : f"R${x/1e6:.1f}M"))

axes[1].plot(od.index, od.values, "s--", color=W_OR, lw=2, ms=5)
axes[1].fill_between(od.index, od.values, alpha=0.08, color=W_OR)
axes[1].set_ylabel("Unique Orders", fontsize=11)
axes[1].set_title("(b) Monthly Order Volume", fontsize=12)
axes[1].yaxis.set_major_formatter(mtt.FuncFormatter(lambda x, _ : f"{int(x):,}"))

axes[2].plot(av.index, av.values, "^-", color=W_GR, lw=2, ms=5)
axes[2].fill_between(av.index, av.values, alpha=0.08, color=W_GR)
axes[2].set_ylabel("Avg Order Value (BRL)", fontsize=11)
axes[2].set_title("(c) Monthly Average Order Value", fontsize=12)
axes[2].yaxis.set_major_formatter(mtt.FuncFormatter(lambda x, _ : f"R${x:,.0f}"))
axes[2].set_xlabel("Month", fontsize=11)

for ax in axes: plt.setp(ax.get_xticklabels(), rotation=45, ha="right")
fig.suptitle("Figure 10. Monthly Revenue, Order Volume, and Average Order Value", fontweight="bold", fontsize=13, y=1.01)
plt.tight_layout(); sv("figure_10.png")
```

# AOV stats

```
aov_start = av.iloc[0]; aov_end = av.iloc[-1]; aov_change = ((aov_end - aov_start) / aov_start) * 100
print(f" AOV start: R${aov_start:.0f}, end: R${aov_end:.0f}, change: {aov_change:+.1f}%")
```

# ——— FIGURE 11: Moving Average ———

```
print("Fig 11")
ma3 = rv.rolling(3, min_periods=2).mean()
fig, ax = plt.subplots(figsize=(11, 5))
ax.plot(rv.index, rv.values, "-", color=W_BL, lw=1.5, alpha=0.5, label="Actual Revenue")
ax.plot(ma3.index, ma3.values, "-", color=W_OR, lw=3, label="3-Month Moving Average")
ax.set_xlabel("Month", fontsize=11); ax.set_ylabel("Revenue (BRL)", fontsize=11)
ax.set_title("Figure 11. Revenue with 3-Month Moving Average")
ax.yaxis.set_major_formatter(mtt.FuncFormatter(lambda x, _ : f"R${x/1e6:.1f}M"))
ax.legend(fontsize=10); plt.xticks(rotation=45, ha="right"); plt.tight_layout(); sv("figure_11.png")
```

```
# ——— FORECASTING ———
```

```
rva=rv.values.astype(float);nm=len(rva);sp=int(nm*0.8)
trn,tst=rva[:sp],rva[sp:];ti,tsi=rv.index[:sp],rv.index[sp:]
print(f"\nForecast: train={len(trn)}, test={len(tst)}")
```

```
def naive(tr,s):return np.array([tr[-1]]*s)
def mafc(tr,s,w=3):return np.array([tr[-w:].mean()]*s)
def sesfc(tr,s,a=0.35):
    v=tr[0]
    for t in tr[1:]:v=a*t+(1-a)*v
    return np.array([v]*s)
def holtfc(tr,s,a=0.4,b=0.15):
    v,bt=tr[0],tr[1]-tr[0]
    for t in tr[1:]:v0,b0=v,bt;v=a*t+(1-a)*(v0+b0);bt=b*(v-v0)+(1-b)*b0
    return np.array([v+(h+1)*bt for h in range(s)])
```

```
# ARIMA Forecast
```

```
arima_model = ARIMA(trn, order=(0,1,1))
arima_fit = arima_model.fit()
```

```
arima_pred = arima_fit.forecast(steps=len(tst))
```

```
preds={
    "Naïve":naive(trn,len(tst)),
    "Moving Avg":mafc(trn,len(tst)),
    "Exp. Smoothing":sesfc(trn,len(tst)),
    "Holt's Linear":holtfc(trn,len(tst)),
    "ARIMA (0,1,1)":arima_pred
}
```

```
def met(a,p):
    return np.mean(np.abs(a-p)),np.sqrt(np.mean((a-p)**2)),np.mean(np.abs((a-p)/(a+1e-9)))*100
mdf=pd.DataFrame({m:met(tst,p) for m,p in preds.items()},index=["MAE","RMSE","MAPE"]).T
best=mdf.RMSE.idxmin()
rmse_naive=mdf.loc["Naïve","RMSE"];rmse_best=mdf.loc[best,"RMSE"]
pct_improv=(1-rmse_best/rmse_naive)*100
print(mdf.round(0))
print(f" Best: {best}, RMSE improvement: {pct_improv:.0f}%")
```

```
# ——— FIGURE 12: Forecast Comparison (bigger) ———
```

```
print("Fig 12")
ms={
    "Naïve":("gray","-", "D"),
```

```

    "Moving Avg":(W_OR,"--","s"),
    "Exp. Smoothing":(W_GR,"-","^"),
    "Holt's Linear":(W_PU,".",None),
    "ARIMA (0,1,1)":(W_VR,"-","o")
}

fig,ax=plt.subplots(figsize=(13,6))
ax.plot(ti,trn,"-",color=W_BL,lw=2.5,label="Actual (Train)")
ax.plot(tsi,tst,"-",color=W_SB,lw=2.5,label="Actual (Test)")
for name,pred in preds.items():
    col,ls,mk=ms[name]
    ax.plot(tsi,pred,ls=ls,color=col,lw=2,marker=mk,ms=5 if mk else 0,label=name)
ax.axvline(tsi[0],color="gray",ls=":",lw=1.2)
ax.set_xlabel("Month",fontsize=11);ax.set_ylabel("Revenue (BRL)",fontsize=11)
ax.set_title("Figure 12. Forecast Comparison: Actual vs. Predicted Monthly Revenue")
ax.yaxis.set_major_formatter(mt.FuncFormatter(lambda x,_:f"R${x/1e6:.1f}M"))
ax.legend(fontsize=10);plt.xticks(rotation=45,ha="right");plt.tight_layout();sv("figure_12.png")

# ——— FIGURE 13: Accuracy Metrics (bigger) ———

print("Fig 13")
mc={
    "Naïve":"gray",
    "Moving Avg":W_OR,
    "Exp. Smoothing":W_GR,
    "Holt's Linear":W_PU,
    "ARIMA (0,1,1)":W_VR
}

fig,axes=plt.subplots(1,3,figsize=(15,5))
for ax,metric in zip(axes,["MAE","RMSE","MAPE"]):
    vals=mdf[metric];bm=vals.idxmin()
    bars=ax.bar(range(len(vals)),vals.values,color=[mc[m] for m in vals.index],edgecolor="black",lw=0.7,width=0.55)
    for i,(bb,v) in enumerate(zip(bars,vals.values)):
        if vals.index[i]==bm:bb.set_edgecolor("black");bb.set_linewidth(3)
        lbl=f"{v:.1f}%" if metric=="MAPE" else f"R${v/1000:.0f}K"
        ax.text(bb.get_x()+bb.get_width()/2,bb.get_height()*1.01,lbl,ha="center",fontsize=9)
    ax.set_xticks(range(len(vals)));ax.set_xticklabels(vals.index,rotation=18,ha="right",fontsize=9)
    ax.set_title(metric,fontweight="bold",fontsize=12)
fig.suptitle("Figure 13. Forecast Accuracy by Model (Lower = Better; Bold Border = Best)",fontsize=12,fontweight="bold",y=1.02)
plt.tight_layout();sv("figure_13.png")

# ——— SAVE ALL VERIFIED STATS ———

stats_out={
    "total_rows":TOTAL_ROWS,"unique_orders":UNIQUE_ORDERS,

```

```

"n_clustering":N_CLUST,
"review_mean":round(rm,2),"review_median":int(rmd),
"r_score_days":round(r_score_days,2),"r_score_delay":round(r_score_delay,2),
"r_days_delay":round(r_days_delay,2),"r_price_pay":round(r_price_pay,2),
"pct_early":round(pct_early,1),"pct_ontime":round(pct_ontime,1),"pct_late":round(pct_late,1),
"sil_k4":round(sil4,3),"sil_k6":round(sil6,3),"sil_diff":round(sil_diff,3),
"K":OK,"anova_f":round(fstat,1),
"pc1_var":round(ev[0]*100,1),"pc2_var":round(ev[1]*100,1),
"aov_start":round(float(aov_start),0),"aov_end":round(float(aov_end),0),
"aov_change_pct":round(aov_change,1),
"train_months":len(trn),"test_months":len(tst),
"best_model":best,
"best_mae":round(float(mdf.loc[best,"MAE"]),0),
"best_rmse":round(float(mdf.loc[best,"RMSE"]),0),
"best_mape":round(float(mdf.loc[best,"MAPE"]),1),
"rmse_improvement_pct":round(pct_improv,0),
"cluster_sizes":{int(k):int(v) for k,v in csizes.items()},
"cluster_pcts":{int(k):round(v/csizes.sum()*100,1) for k,v in csizes.items()},
}

with open(os.path.join(OUT,"_verified_stats.json"),"w") as f:
    json.dump(stats_out,f,indent=2)

mdf.round(1).to_csv(os.path.join(OUT,"_forecast_metrics.csv"))
cmeans.round(2).to_csv(os.path.join(OUT,"_cluster_means.csv"))
audit_df.to_csv(os.path.join(OUT,"_join_audit.csv"),index=False)

saved=sorted([f for f in os.listdir(OUT) if f.endswith(".png")])
print(f"\n{' '*50}\n{len(saved)} figures saved\n{' '*50}")
print(json.dumps(stats_out,indent=2))

```